FLOOD RISK ANALYSIS AND VISUALIZATION WEB PORTAL FOR LAGOS STATE, NIGERIA

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A PROJECT REPORT

SUBMITTED TO



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ABSTRACT

Flooding is the most frequent and costly natural hazard affecting Lagos State, yet flood-risk information for the state remains fragmented, static, and difficult for non-specialist stakeholders to use. This study designed and built an interactive, web-based flood-risk analysis and visualization portal for Lagos State, integrating open geospatial and hydro-meteorological datasets with a process-based hydrological model and a multi-criteria susceptibility assessment. A SWAT+ model of the Ogun catchment (delineated area 19,915 km², outlet at Kara/Akute) was forced with CHIRPS rainfall and ERA5-Land temperature and calibrated against GEOGLOWS version 2 retrospective discharge over 2011-2017, with validation over 2018-2023, gauged NIHSA records being unavailable within the project period. The calibrated model met all three performance criteria over the full evaluation period (Nash-Sutcliffe Efficiency 0.507, R-squared 0.606, Percent Bias +1.6%), with validation-period performance exceeding calibration-period performance; an independent evapotranspiration check against MODIS and TerraClimate reproduced annual totals to within 14.4% while underestimating seasonal amplitude. A statewide flood-susceptibility map was produced by AHP-weighted overlay of eight conditioning factors (consistency ratio 0.0115) and classified into five zones. Validated against an independent MODIS 2012 flood extent, the map ranked susceptibility correctly - enrichment rose monotonically to 1.61 times chance in the Very High zone, and 67.08% of observed flooding fell within the High and Very High zones - but met only one of its four discrimination criteria (AUC-ROC 0.62, sensitivity 0.685, overall accuracy 0.526), reflecting a deliberately strict, independent validation and the near-half of the state occupying the upper zones. An estimated 4,810,069 residents, or 38.7% of the population, were found to occupy High or Very High susceptibility zones, ranging from 67.8% in Eti-Osa to 7.2% in Ifako/Ijaye. The outputs were delivered through a lightweight Streamlit portal providing hazard, exposure, scenario, hydrology and validation views. The study demonstrates that credible, transparent, decision-support flood products can be assembled for a data-scarce megacity from open data, while honestly delimiting where such products rank rather than classify flood risk.

CERTIFICATION

This is to certify that this project was carried out by Attah Gbubemi David with Matriculation number 190402060 of the Department of Civil and Environmental Engineering, Faculty of Engineering, University of Lagos, in partial fulfilment of the requirements for the award of Bachelor of Science (B.Sc.) degree in Civil and Environmental Engineering.

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DEDICATION

This project is dedicated to God Almighty, whose grace, wisdom, and endless mercy carried me through every stage of this journey.

And to my dear mother — my first teacher, my constant source of strength, and my greatest inspiration. Your love, sacrifice, and unwavering belief in me made this work possible.

ABBREVIATIONS

AHP Analytical Hierarchy Process

AUC-ROC Area Under the Receiver Operating Characteristic Curve

AWS Amazon Web Services

CHIRPS Climate Hazards Group InfraRed Precipitation with Station Data

CN / SCS-CN Curve Number / Soil Conservation Service Curve Number

CRS Coordinate Reference System

DEM Digital Elevation Model

DFO Dartmouth Flood Observatory

ERA5-Land Fifth-generation ECMWF Reanalysis, Land component

ESA European Space Agency

ET / PET Evapotranspiration / Potential Evapotranspiration

FSI Flood Susceptibility Index

GEOGLOWS Global Earth Observation for Integrated Water Resource Assessment streamflow service

GIS Geographic Information System

GLO-30 Copernicus Global 30 m Digital Elevation Model

HAND Height Above Nearest Drainage

HDX Humanitarian Data Exchange

HRU Hydrologic Response Unit

HSG Hydrological Soil Group

LASEMA Lagos State Emergency Management Agency

LGA Local Government Area

MCDA Multi-Criteria Decision Analysis

MODIS Moderate Resolution Imaging Spectroradiometer

NEMA National Emergency Management Agency

NIHSA Nigeria Hydrological Services Agency

NiMet Nigerian Meteorological Agency

NSE Nash-Sutcliffe Efficiency

PBIAS Percent Bias

ROC Receiver Operating Characteristic

SRTM Shuttle Radar Topography Mission

SUFI-2 Sequential Uncertainty Fitting version 2

SWAT / SWAT+ Soil and Water Assessment Tool

TWI Topographic Wetness Index

WorldPop WorldPop gridded population product

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Flooding is one of the most destructive and recurrent natural hazards confronting urban centres across the globe. Lagos State, Nigeria, is one of the worst-hit metropolitan areas in sub-Saharan Africa when it comes to flooding. Lagos sits on a low-lying coastal plain, hemmed in by the Lagos Lagoon, Lekki Lagoon, Badagry Creek, and the lower reaches of the Ogun River run right through Lagos State (Adelekan, 2020), and with seasonal rainfall regularly topping 1,500 mm a year, the area faces serious risk from both surface water and river flooding. Over the decades, rapid and largely unplanned growth has swallowed up wetlands and farmland, replacing them with hard surfaces that don't absorb water. The drainage system, which was never designed for anything close to today's population, simply can't keep up (Nkwunonwo *et al*., 2020).

With a population now above 15 million by recent estimates, Lagos is Africa's largest city, and it's still growing faster than the infrastructure needed to support it. Because drainage capacity has lagged so far behind urban expansion, even a moderate downpour can leave large parts of the city under water (Adeleye and Popoola, 2019). That reality has played out repeatedly: the July 2011 floods, the back-to-back inundations of 2017 and 2020, and the catastrophic 2022 flooding that displaced over 1.4 million people across Nigeria and hit Lagos hard all point to a clear need for better flood risk planning (Echendu, 2022; NIHSA, 2022). Year after year, the Lagos State Emergency Management Agency ranks flooding as the most frequent and expensive natural hazard the state faces, with direct losses climbing into the billions of naira annually (LASEMA, 2023).



Figure 1: Flooding in Lagos disrupts transportation and forces residents to navigate deeply inundated streets (The Nation, 2019)

Despite how severe and frequent flooding is in Lagos, a persistent challenge identified during preliminary research was the difficulty to find integrated, regularly updated flood risk information. Most flood studies come out as one-off outputs from individual research projects or agencies, stored in formats that require specialized GIS skills to use (Oladokun and Proverbs, 2016). There are useful datasets on rainfall, terrain, soil, land cover, and river networks at both national and global scales, but they are almost never brought together into one platform that non-specialist users like urban planners, emergency managers, and local government officials can easily pull up and explore on their own. Most flood mapping work done in Lagos has also been static, offering a single snapshot of risk without any way to compare different rainfall scenarios or update the results as new data comes in (Nkwunonwo *et al*., 2020; Olawuni *et al*., 2023).

This gap motivated the development of a web-based flood risk analysis and visualization portal that pulls together hydro-meteorological, terrain, land cover, soil, and river datasets, supports mapping workflows that are repeatable and transparent, and presents the analytical outputs through interactive maps and dashboards that a wide range of stakeholders can actually understand. Such a portal would move flood risk information from static reports and specialist GIS desktops into an interactive, updatable, and shareable digital environment. By incorporating satellite-derived rainfall from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), terrain and

drainage derivatives from the Copernicus GLO-30 Digital Elevation Model (DEM), river and basin delineations from the HydroSHEDS suite (HydroRIVERS and HydroBASINS), land cover patterns from the European Space Agency (ESA) WorldCover product, and soil hydraulic properties from SoilGrids, the portal would rest on a foundation of openly accessible and scientifically documented global datasets. Where Nigerian Meteorological Agency (NiMet) rain gauge records and Nigeria Hydrological Services Agency (NIHSA) discharge or water level data are available, they would be used to cross-check and calibrate the results, strengthening local relevance (NiMet, 2023; NIHSA, 2022).

In this study, hydrological simulation was also integrated into the portal using the Soil and Water Assessment Tool (SWAT or SWAT+), a physically based, semi-distributed watershed model that has been widely applied in Nigerian river basins for runoff estimation and scenario analysis (Aduah *et al*., 2020). SWAT-based simulation would provide quantitative estimates of runoff depth and streamflow under baseline and user-defined scenario rainfall conditions, thereby introducing a dynamic, process-based dimension to what would otherwise be a purely index-based susceptibility mapping exercise. The susceptibility and risk mapping itself would follow a transparent GIS multi-criteria decision analysis (MCDA) approach using the Analytical Hierarchy Process (AHP), a widely adopted multi-criteria weighting technique (Saaty, 1980; Swain *et al*., 2020; Nahin *et al*., 2023), to weight factors such as elevation, slope, flow accumulation, distance to rivers, rainfall intensity, land cover type, soil infiltration capacity, and SWAT-derived runoff indicators. The resulting layers would then be combined with exposure data on population, buildings, and roads to produce risk maps. Users can explore these interactively through the portal, filter by Local Government Area (LGA) or sub-basin, and export results for reporting. The goal is to make the outputs immediately useful, not just technically sound.

Put simply, this study was undertaken because Lagos State critically requires a single, transparent, and easy-to-use platform for flood risk analysis, one that pulls together the best available data, supports hydrological modelling and multi-criteria assessment, and puts the results in front of

decision-makers and the public through modern web-based visuals and dashboards. The remaining sections of this chapter present the specific problem statement, aim and objectives, significance, study area description, scope, limitations, key definitions, and chapter organisation for the study.

The study focuses on Lagos State, whose geographic and climatic characteristics are described in the following paragraphs.

Lagos State is located in the southwestern corner of Nigeria between approximately latitudes 6.4 degrees North and 6.7 degrees North and longitudes 3.1 degrees East and 4.0 degrees East. The state occupies a total land area of approximately 3,577 square kilometres, of which a significant proportion consists of water bodies including lagoons, creeks, and wetlands. To the south, Lagos is bounded by the Atlantic Ocean; to the north and east by Ogun State; and to the west by the Republic of Benin. The topography is predominantly flat and low-lying, with much of the urban area situated at elevations below 5 metres above mean sea level, making it one of the most flood-vulnerable coastal cities in Africa (Adelekan, 2020).



Figure 2: Map of Lagos State, Nigeria (Auwalu and Bello, 2023)

in areas adjacent to the lagoon shoreline and along the lower reaches. The drainage system in Lagos revolves around the Lagos Lagoon, which takes in freshwater from the Ogun River, the Majidun River, and a number of smaller streams. To the east sits the Lekki Lagoon, and to the west, the Badagry Creek. These linked water bodies matter ecologically, but they also limit how quickly stormwater can drain out, especially during heavy rainfall when tides in the Atlantic are high. The backwater effect this creates makes flooding worse in areas close to the lagoon shoreline and along the lower reaches of inflowing rivers (Obiefuna *et al*., 2021).

The climate of Lagos is tropical, which in practical terms means it rains heavily for about seven months of the year (April to October) and is relatively dry for the other five (November to March). During the rainy season, roads flood within minutes of a heavy downpour, traffic grinds to a halt, and areas near the lagoon can remain waterlogged for days. Mean annual rainfall ranges from approximately 1,500 mm to over 2,000 mm, with the heaviest rains typically occurring between June and July. The principal NiMet synoptic weather station for the Lagos area is located at the Murtala Muhammed International Airport, Ikeja (latitude 6.58 degrees North, longitude 3.33 degrees East, elevation approximately 38 m above sea level). A secondary station, the Lagos Roof station, is also operated within the metropolitan area. Together, these stations provide the ground-based rainfall records used for validating the CHIRPS satellite rainfall estimates in this study. Short-duration, high-intensity rainstorms are common and frequently overwhelm the capacity of the urban drainage system, which was designed for a much smaller and less densely built-up city (Adeleye and Popoola, 2019).

Lagos has urbanized at a remarkable pace over the past five decades. The population has jumped from under 1.5 million in 1970 to over 15 million by recent estimates, and the built-up area has spread to match, swallowing up wetlands, farmlands, and open spaces and replacing them with hard, impervious surfaces. This land-use transformation has profoundly altered the hydrological

response of the landscape, increasing surface runoff volumes and peak flows, reducing lag times, and overwhelming drainage channels that suffer from inadequate capacity, poor maintenance, and blockage by solid waste (Obiefuna *et al*., 2021; Olawuni *et al*., 2023). Places like Lekki, Ikoyi, Victoria Island, Ajegunle, Agege, Ikorodu, Ojo, Alimosho, Surulere, and Mushin come up again and again as some of the worst-hit areas, though in reality flooding touches virtually every local government area to some extent (LASEMA, 2023).

In this study, the web portal is designed to cover the entirety of Lagos State, providing statewide flood susceptibility and risk visualization. However, the hydrological modelling component using SWAT or SWAT+ focuses on one or more representative catchments draining into the Lagos Lagoon system, selected on the basis of data availability, manageable spatial extent, and hydrological significance. This approach balances comprehensive coverage in the portal interface with the practical feasibility constraints of physically-based watershed modelling in a data-limited environment.

1.2 Statement of the Problem

Lagos State keeps getting hit by urban floods that are growing more intense and more damaging with each passing year. According to a December 2024 joint assessment by the Displacement Tracking Matrix and NEMA, flooding in Lagos affected 275,621 individuals across 48,403 households in 14 Local Government Areas, with 2,313 persons displaced and 7% of affected houses totally destroyed (DTM, 2024). Nationally, the 2024 floods killed at least 285 people and displaced over 641,000 across 31 states (OCHA, 2024). Every year, flooding affects around 90,000 Lagos residents. These events have repeatedly disrupted everyday life across the city. Available research indicates that the July 2021 floods alone caused an estimated 380 billion Naira (US$320 million) in property damage across Lagos, with approximately 100 fatalities (Oladunjoye, 2011; Higuera Roa *et al*., 2022). The 2022-2023 flood events resulted in US$262,500 in direct damages and displaced 8,000 residents in Lagos State (Frontiers in Climate, 2025). These numbers paint a clear picture: the

annual cost of flood damage across the Lagos metropolis runs into tens of billions of naira, and informal and low-income settlements take the worst of it (Adelekan, 2020; Echendu, 2022). Government agencies including LASEMA, NiMet, and NIHSA generate various forms of flood-related data and early warning products, while academic researchers have produced numerous flood susceptibility and hazard maps for parts of Lagos using remote sensing and GIS techniques (Agbola *et al*., 2012; Nkwunonwo *et al*., 2020; Olawuni *et al*., 2023). But despite all of this, several serious gaps remain.

The first issue is simple but stubbornly persistent: flood risk information for Lagos is spread across individual studies, agency databases, and project reports. There is no single platform pulling together terrain, rainfall, soil, land cover, river network, and hydrological modelling outputs into one unified, explorable analytical environment. Second, most existing flood maps are made using desktop GIS software and come out as static image files or PDF reports. You need specialist knowledge to reinterpret or update them. Non-specialist stakeholders, whether urban planners at the local government level, emergency response coordinators, community groups, or real estate professionals, generally do not have the software, the skills, or the time to engage meaningfully with these outputs (Oladokun and Proverbs, 2016). Third, there is no easy way to compare flood risk under different rainfall scenarios. If you wanted to compare, say, a baseline annual rainfall pattern with one associated with an extreme event or a projected climate shift, there is no tool for doing that. And there is no mechanism for refreshing the analyses when new rainfall data come in. This means that flood risk assessments in Lagos tend to be one-time exercises rather than living, updatable decision-support tools.

The problem addressed in this study is the absence of an integrated, interactive, and updatable web-based flood risk analysis and visualization system for Lagos State that brings together the relevant datasets, supports transparent and repeatable susceptibility and risk mapping, incorporates process-based hydrological simulation for scenario exploration, and presents results through dashboards and map interfaces accessible to non-specialist users. Without such a system, flood risk science cannot

easily translate into practical planning and emergency response in Lagos. That is the core challenge addressed in this project.

1.3 Aim and Objectives

The aim of this study is to design and build an interactive web GIS portal for flood risk analysis and visualization in Lagos State, Nigeria, integrating multi-source geospatial and hydro-meteorological datasets and supported by SWAT-based hydrological simulation, to provide accessible, repeatable, and scenario-capable flood risk information for a broad range of stakeholders.

The specific objectives of the study are as follows:

(i) To acquire, preprocess, and harmonise at least six geospatial datasets (CHIRPS daily rainfall, Copernicus GLO-30 DEM, HydroRIVERS river and basin layers, ESA WorldCover land cover, SoilGrids soil properties, and available NiMet records, with GEOGLOWS retrospective discharge as the calibration reference where gauged records prove unavailable) into a spatially consistent, analysis-ready data stack covering Lagos State at 30 m resolution in the EPSG:32631 projected coordinate system. Success criterion: all layers clipped, reprojected, and resampled with no missing coverage over the study area.

(ii) To set up and run a SWAT+ hydrological model for a representative catchment draining into the Lagos Lagoon system, achieving calibration and validation performance of Nash-Sutcliffe Efficiency (NSE) greater than or equal to 0.50, R-squared greater than or equal to 0.60, and Percent Bias (PBIAS) within plus or minus 25%, and to produce runoff and streamflow outputs under at least three rainfall scenarios (baseline, extreme historical event, and climate sensitivity).

(iii) To produce flood susceptibility and risk maps for Lagos State using a GIS-based multi-criteria decision analysis (MCDA) approach with Analytical Hierarchy Process (AHP) weighting (consistency ratio of 0.10 or less), integrating eight conditioning factors, and to validate the maps against an independent flood inventory, reporting AUC-ROC, sensitivity, specificity and overall

accuracy against the acceptance criteria of Table 3.6, with an AUC-ROC above 0.70 taken as the threshold for acceptable discrimination.

(iv) To develop a functional interactive web portal providing at least five core features (layer control with legends, a time-series dashboard, a scenario runner, spatial query popups, and downloadable outputs) built around an interactive web map, implemented on a lightweight architecture appropriate to the static raster and tabular nature of the outputs.

(v) To validate all modelling and mapping outputs against the defined acceptance thresholds of Table 3.6 (SWAT+ model NSE greater than or equal to 0.50; susceptibility-map AUC-ROC above 0.70; and more than 60% of observed flood extent falling within the High or Very High susceptibility zones), reporting each outcome as obtained. Validation data comprise an independent satellite-derived flood extent, agency flood records, and NIHSA's Annual Flood Outlook.

1.4 Significance of the Study

This study has the potential to close the long-standing gap between flood risk science and actual decision-making in Lagos State. A lot of academic and institutional work has gone into understanding flood dynamics in Lagos, but the results of that work too often stay locked inside technical reports and journal papers that the people who need them most cannot easily get to or use (Nkwunonwo *et al*., 2020; Adeleye and Popoola, 2019). By putting flood risk information into an interactive web portal with map exploration, dashboards, and scenario comparison tools, this study goes directly at the accessibility problem that has limited how useful existing flood assessments are.

For government agencies like LASEMA, the Lagos State Ministry of the Environment, the Lagos State Urban Renewal Authority, and local government councils, the portal would provide a single, integrated view of flood susceptibility across the state. Consultations with LASEMA officials during preliminary research revealed a recurring frustration with having to piece together information from different sources before flood response could begin. The portal was designed to address this problem. Emergency management staff would be able to spot high-risk zones quickly

and communicate them to first responders and affected communities in a visual format that is easy to read. The time-series dashboards for rainfall and discharge would help track hydro-meteorological trends that might signal higher flood risk, and the scenario runner would let planners explore "what-if" conditions without requiring GIS expertise.

On the academic and research side, It is hoped that this study demonstrates, in a practical way, that openly available global datasets can be combined with a process-based hydrological model and a multi-criteria geospatial framework to produce flood risk products that are actually useful in a data-sparse tropical urban setting. Because the methodology is transparent and repeatable, with all spatial processing done in QGIS and all data sources clearly documented, it should be possible to replicate and adapt this work for other Nigerian cities dealing with similar flood problems, including Ibadan, Port Harcourt, and Benin City (Aderogba, 2012).

The study also connects with several national and international policy priorities. It directly supports SDG 11 (Sustainable Cities and Communities), specifically Target 11.5 which aims to significantly reduce the number of deaths and people affected by disasters including floods, and Target 11.b which calls for cities to adopt integrated disaster risk management strategies in line with the Sendai Framework (UN, 2015). The study further supports SDG 13 (Climate Action), particularly Target 13.1 on strengthening resilience and adaptive capacity to climate-related hazards, and Target 13.3 on improving education and awareness of climate change impacts. It is also relevant to SDG 1 (No Poverty), Target 1.5, which focuses on building the resilience of the poor and vulnerable to climate-related extreme events, and SDG 6 (Clean Water and Sanitation), Target 6.5, on integrated water resources management. At the national level, the study aligns with Nigeria's National Flood Emergency Preparedness and Response Plan, which calls for improved flood forecasting and early warning capabilities (NEMA, 2020), and the Sendai Framework for Disaster Risk Reduction (UNDRR, 2015). The portal also has educational value, as it can serve as a teaching resource for university courses in hydrology, environmental management, and urban planning, as well as for community-level flood awareness campaigns.

1.5 Scope and Limitations of the Study

The scope of this study covers the following:

Geographic scope: The web portal and the associated flood susceptibility and risk maps cover the full extent of Lagos State, encompassing all twenty Local Government Areas. The SWAT or SWAT+ hydrological modelling, which requires detailed catchment setup and calibration, is limited to one or more representative catchments within or draining into the Lagos Lagoon system, selected for feasibility and data availability.

Technical scope: Spatial data preprocessing, terrain analysis, susceptibility and risk mapping, and map layer preparation are carried out in QGIS. Hydrological simulation is performed using SWAT or SWAT+ with the data inputs described, including CHIRPS rainfall, Copernicus GLO-30 DEM, ESA WorldCover land cover, SoilGrids soil parameters, and HydroSHEDS river and basin products. The multi-criteria susceptibility mapping follows an AHP-weighted overlay approach integrating elevation, slope, flow accumulation, proximity to rivers, land cover class, rainfall extremes, soil infiltration proxy, and SWAT-derived runoff indicators. The web portal provides interactive map layer control with identify tools and legends, time-series dashboards for rainfall and discharge, a scenario runner supporting predefined rainfall events and controlled uploads with input validation, and reporting or export by LGA or sub-basin.

Data scope: The study relies primarily on openly available global and regional datasets. Local gauge rainfall from NiMet and discharge or water level records from NIHSA are used where available for bias-checking and calibration or validation, though the availability of such records is acknowledged as constrained. Exposure layers representing population distribution, buildings, and road networks are incorporated where licensing and availability permit, with constraints clearly noted.

Exclusions: The study does not undertake real-time flood forecasting or nowcasting. It does not include detailed hydraulic modelling of urban drainage networks or storm sewers at the pipe level.

Coastal and tidal surge modelling is outside the scope. Climate change projection modelling under specific Representative Concentration Pathway (RCP) or Shared Socioeconomic Pathway (SSP) scenarios is not performed, though the scenario runner allows exploration of different rainfall magnitudes. Ground-truth survey of flood depths at property level is not conducted; rather, validation relies on published flood evidence and agency reports.

Beyond these boundaries, the study has a number of practical limitations that are laid out clearly below, along with how they were addressed.

The single biggest limitation of this project is the gap between what the methodology promises on paper and what the data can actually support. Lagos is frequently described as data-scarce, a characterisation confirmed during this research. Discharge records that were supposed to cover 20 years had gaps of 3 to 5 years in them. Some NiMet rainfall files turned up with months missing for no clear reason. The SoilGrids data, while globally consistent, may not capture the highly localised fill material and compacted surfaces that define much of Lagos's built environment. These problems were addressed as effectively as possible, though no amount of methodological care fully compensates for imperfect input data, and this limitation is acknowledged transparently.

Gauge and discharge data scarcity: Ground-based meteorological records for the Lagos area were obtained from the Nigerian Meteorological Agency (NiMet) for the Ikeja synoptic station as a continuous 30-year annual series (1996–2025). This record is used as the ground-based reference against which the CHIRPS satellite rainfall is validated, and as the basis for the design-rainfall return-period analysis reported in Chapter Four; the gridded CHIRPS product supplies the statewide spatial rainfall surfaces. The principal remaining data constraint is therefore on the discharge side: NIHSA river discharge records for streams within the Lagos Lagoon basin are sparse and limited in temporal coverage, which constrains the robustness of SWAT model calibration and validation. To manage this, the study uses CHIRPS satellite rainfall for spatial and long-term coverage — validated against the NiMet Ikeja gauge record and shown to perform well in West African contexts

(Funk *et al*., 2015) — and applies accepted SWAT performance-evaluation metrics (Moriasi *et al*., 2007) alongside qualitative spatial validation against published flood evidence.

DEM accuracy in flat terrain: The Copernicus GLO-30 DEM, though an improvement on the earlier SRTM product, still carries vertical uncertainty of a few metres in areas of low relief, and the flat coastal topography of Lagos means that small elevation errors can significantly affect the delineation of flow paths and flood-prone depressions. This uncertainty is acknowledged and discussed in the interpretation of results. Terrain derivatives are cross-checked against HydroSHEDS river lines and known drainage patterns.

Exposure data licensing and coverage: High-resolution datasets describing building footprints, population distribution at fine scales, and road networks may be subject to licensing restrictions or may not be uniformly available across all parts of Lagos. The study uses the best available exposure layers and clearly documents their sources, resolution, and limitations. Where datasets are unavailable, exposure assessment is described at a coarser level or acknowledged as a gap.

Portal computational constraints: The web portal is designed as a prototype demonstrating the feasibility of interactive flood risk visualization and scenario comparison. Its computational capacity for processing very large or real-time datasets may be limited relative to fully operational enterprise-level systems. The study documents these constraints transparently and discusses pathways for scaling.

Subjectivity in AHP weighting: This represents one of the most challenging aspects of the project. The AHP method requires the analyst to make judgement calls about which factors matter more than others, and different experts could reasonably arrive at different weights (Saaty, 1980; Nahin *et al*., 2023). Consistency ratio checks are applied as prescribed in the AHP framework, and sensitivity to weight choices is discussed.

Model structural uncertainty: SWAT is a semi-distributed, conceptual-to-physical model that simplifies complex hydrological processes. Its application in a highly urbanised, flat, tidally

influenced coastal environment such as Lagos presents particular challenges, including the representation of urban stormwater routing and lagoon backwater effects. These structural limitations are acknowledged, and results are interpreted with appropriate caution.

No gauged discharge. The model was calibrated and validated against modelled GEOGLOWS discharge rather than gauged records, because NIHSA data could not be obtained within the project period. The reported performance statistics therefore measure inter-model agreement rather than agreement with observation. This is partly offset by the independent evapotranspiration validation (Chapter Four), but it remains a genuine constraint on the strength of the hydrological claim.

Oyan Dam not represented. Oyan Dam lies within the modelled Ogun basin, and its releases are not represented in the model. The 2022 flood was substantially release-driven and therefore falls outside the model's representational scope, as shown by the near-baseline discharge simulated for the 2022 rainfall in Chapter Four.

Single-layer soil. Soils were represented using only the SoilGrids 0-5 cm topsoil texture, with approximately 88% of the basin assigned to a single texture class and no vertical soil structure resolved. This simplifies infiltration and percolation behaviour in a way that the calibration can only partly compensate for.

Single-event, coarse-resolution spatial validation. The susceptibility map is a static surface tested against a single flood season (the MODIS 2012 extent) at 250 m resolution, which cannot resolve street-scale urban flooding; non-detection at that resolution is therefore not evidence of no flooding.

No uncertainty ensemble. Because calibration was performed by direct parameter adjustment rather than SUFI-2, the P-factor and R-factor uncertainty statistics were not produced. Generating them through a subsequent SUFI-2 run is identified as further work.

CHAPTER TWO

LITERATURE REVIEW

This chapter reviews the existing literature that is relevant to building a Flood Risk Analysis and Visualization Web Portal for Lagos State, Nigeria. It synthesizes (i) Lagos-specific flood drivers, impacts, and evidence base; (ii) transparent Geographic Information System (GIS)–Multi-Criteria Decision Analysis (MCDA)/Analytical Hierarchy Process (AHP) approaches for flood susceptibility and risk mapping; (iii) the role of the Soil and Water Assessment Tool (SWAT)/SWAT+ hydrological simulation in producing runoff/streamflow indicators and supporting scenario analysis; (iv) the major open datasets underpinning a repeatable workflow (Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), Shuttle Radar Topography Mission (SRTM), HydroSHEDS/HydroRIVERS, European Space Agency (ESA) WorldCover, SoilGrids, and local Nigerian agency data where accessible); and (v) literature on web GIS flood platforms and standards for publishing interactive flood-risk information to non-specialist users.

The literature clearly establishes that flooding in Lagos State results from the interaction of hydrometeorological forcing, low-lying coastal terrain, and rapid urban transformation, with risk made worse by high exposure and uneven vulnerability. Researchers have responded by turning increasingly to spatially explicit tools, among them GIS-based flood susceptibility mapping, remote-sensing land cover characterization, hydrological modelling, and risk communication products, to support decision-making (Nkwunonwo *et al*., 2016; Adelekan, 2016; Nwogu *et al*., 2025). This chapter therefore reviews the knowledge base that supports an integrated workflow in which repeatable GIS analysis (QGIS-based) and hydrological simulation (SWAT/SWAT+) are combined with transparent MCDA/AHP to generate interpretable flood susceptibility and risk layers, and then disseminated through a web portal designed for accessibility, scenario exploration, and updates.

2.1 Flooding in Lagos: Drivers, Impacts, and Local Evidence

Lagos State has been recognised for a long time as one of the most flood-prone coastal urban areas in Nigeria. Natural and human factors come together here in ways that make flooding both frequent and unevenly distributed across the city (Adelekan, 2010; Adelekan, 2016; Nkwunonwo *et al*., 2016). Physically, Lagos occupies a barrier-lagoon coastal setting characterized by extensive water bodies and wetlands, low relief, and shallow groundwater conditions that reduce natural storage and drainage efficiency. These conditions shape a flood regime in which pluvial (urban stormwater) flooding is common during intense rainfall, while fluvial and coastal/tidal influences interact with lagoon and river networks to aggravate inundation in low-lying neighborhoods and floodplains (Adelekan, 2016; Nkwunonwo *et al*., 2016).

One of the defining factors in Lagos is the pace of urbanization and land cover conversion, which has increased impervious surfaces and disrupted the natural drainage pathways the city depends on. Land use change studies in Lagos have highlighted that urban expansion alters surface runoff generation and increases flood susceptibility by replacing vegetation and permeable soils with built-up surfaces and modified drainage infrastructure (Idowu and Zhou, 2021). The erosion of natural buffers is particularly important because wetlands and floodplains that historically acted as storage and attenuation zones are increasingly encroached upon or filled, thereby elevating runoff peaks and reducing the capacity of the landscape to dissipate stormwater (Adelekan, 2016; Idowu and Zhou, 2021). In addition, blocked or undersized drains, channel modifications, and poor maintenance, typical in dense urban environments, can convert even moderate rainfall events into disruptive urban floods (Nkwunonwo *et al*., 2016).

But flooding in Lagos is not just about physical processes. It is also, at its core, a socio-infrastructural problem. Informal settlement growth on marginal land, limited enforcement of planning regulations, and unequal access to resilient infrastructure contribute to uneven flood impacts. Community-level studies show that flood risk is not only a function of hazard but also of

exposure and vulnerability: households with weaker housing materials, limited savings, and constrained access to services may experience disproportionate losses and slower recovery after floods (Adelekan, 2010; Ajibade *et al*., 2013). Risk perception research further indicates that lived experience of floods shapes local understanding of risk and willingness or ability to adopt coping strategies, underscoring the importance of communicating risk in forms that are intelligible and actionable for different user groups (Adelekan and Asiyanbi, 2016). These findings matter directly for the design of a flood-risk web portal. Risk communication has to address not only where flooding happens but the fact that different communities experience that risk very differently.

Flooding in Lagos hits multiple sectors at once: housing and public infrastructure get damaged, transport corridors are disrupted, businesses lose revenue, and there are serious public health concerns tied to contaminated floodwaters and the environmental conditions that follow (Adelekan, 2016; Ajibade *et al*., 2013). Institutional reports also reinforce the operational consequences of Lagos floods and the need for systematic documentation and mapping. For example, a Joint Post-Flood Assessment for Lagos State applied field observation and stakeholder engagement to document impacts and needs, demonstrating the continuing reliance on event-based assessment and the value of integrating such evidence into geospatial risk products for response and preparedness (IOM, 2024). At the state level, flood vulnerability assessment and mapping efforts have been reported by the Lagos State Emergency Management Agency (LASEMA), indicating an institutional recognition that spatial flood risk products are central to planning, early warning, and emergency response (LASEMA, n.d.).

When it comes to mapping, existing efforts for Lagos have used a mix of approaches. Most rely on Digital Elevation Model (DEM)-based terrain derivatives, proximity to waterways, land use/land cover indicators, and rainfall-related variables to identify flood-prone zones (Adeaga *et al*., 2020; Idowu and Zhou, 2021; Isiaka *et al*., 2023). However, the Lagos literature also highlights recurring limitations, including incomplete flood inventories for validation, sensitivity to data resolution in low-relief terrain, and the challenge of translating research outputs into accessible decision-support

tools (Nkwunonwo *et al*., 2016; Nwogu *et al*., 2025). These constraints motivate an approach that combines interpretable modelling with a dissemination platform (web portal) that supports ongoing updates and stakeholder use.

2.2 Flood Susceptibility Mapping Approaches

A notable challenge in the literature is the inconsistency of terminology. Different authors use terms like susceptibility, hazard, vulnerability, and risk in slightly different ways, and the definitions do not always line up from one study to the next. In practice, susceptibility mapping typically represents the relative likelihood that a location is prone to flooding given static or semi-static conditioning factors such as terrain, drainage, land cover, and soils, while hazard mapping often incorporates magnitude or probability elements where hydrological/hydraulic modelling or event frequency information is available (Nkwunonwo *et al*., 2016; Nwogu *et al*., 2025). Risk is more comprehensively conceptualized as a function of hazard interacting with exposure and vulnerability, meaning that a high-susceptibility zone may not necessarily translate to high risk if exposure is low, and conversely a moderate hazard may produce large losses where exposure and vulnerability are high (Adelekan, 2010; Nwogu *et al*., 2025). For Lagos, this conceptual separation is important because flood impacts are strongly shaped by population density, land use, and socio-economic conditions (Ajibade *et al*., 2013; Adelekan and Asiyanbi, 2016).

Within this landscape, GIS-based multi-criteria decision analysis (MCDA) has stayed popular for flood susceptibility mapping. The reason is straightforward: it offers a transparent way of integrating different factors into a single index surface. Among MCDA methods, the Analytical Hierarchy Process (AHP) is frequently adopted to derive weights through pairwise comparison of criteria, allowing experts to encode relative importance in a structured manner while checking for logical consistency (Saaty, 1987). In flood studies, AHP is valued for two reasons that align directly with the needs of a web portal. First, it is interpretable: stakeholders can see which factors drive the final susceptibility or risk surface, and weights can be explained in policy and engineering contexts.

Second, it is operational: once the workflow is defined, the mapping can be repeated when new inputs become available (e.g., updated rainfall or land cover), supporting iterative planning rather than one-off products.

In Nigeria, MCDA/AHP approaches have been used to map flood-prone areas in diverse urban and basin contexts. Ajibade *et al*. (2021) applied a GIS-based analytical hierarchy framework for flood-prone mapping in Ibadan, illustrating how AHP supports structured decision-making where multiple drivers coexist. Similarly, Nkeki *et al*. (2022) proposed an analytic hierarchical raster fusion approach for flood risk mapping in a data-scarce Nigerian basin, emphasizing the practicality of AHP-informed raster integration when observations are limited. Although these studies are not Lagos-specific, they are methodologically relevant because they highlight the trade-offs inherent to MCDA in Nigeria: interpretability and feasibility under limited data, balanced against subjectivity in weight assignment and sensitivity to input resolution.

In Lagos-focused applications, susceptibility and hazard mapping studies often integrate classic conditioning factors, including elevation, slope, distance to drainage, flow accumulation indicators, land cover, and rainfall-related variables, into weighted overlay or hybrid frameworks. For example, Idowu and Zhou (2021) explicitly situate flood hazard mapping within the context of land use/land cover change in Lagos State, reflecting the central role of urban expansion and wetland alteration in shaping susceptibility. Lagos studies have also used data-driven alternatives such as the frequency ratio model to estimate susceptibility patterns from conditioning factors (Isiaka *et al*., 2023). While such data-driven methods may reduce reliance on expert judgement, they still require careful validation and may depend on the availability and quality of flood inventory data, which remains a persistent limitation in many Nigerian settings (Nkwunonwo *et al*., 2016; Nwogu *et al*., 2025). For a portal designed for public-sector planning and non-specialist communication, AHP/MCDA retains a clear advantage: it can be explained, audited, and updated without requiring opaque model structures.

A rigorous MCDA/AHP framework requires not only factor selection and weighting but also attention to standardization, classification, and uncertainty. In Lagos, where topographic gradients are small, susceptibility mapping can be especially sensitive to DEM error, hydrological conditioning choices, and the resolution at which drainage networks are represented (Nkwunonwo *et al*., 2016). This sensitivity implies that a transparent workflow, where assumptions are documented and where users can interrogate factor layers in a portal, is as important as the final susceptibility index itself.

2.3 Hydrological Simulation and the Role of SWAT/SWAT+ in Flood Studies

MCDA susceptibility mapping gives a useful picture of flood-prone conditions, but it has a core limitation: it does not capture the dynamic process by which rainfall becomes runoff and channel flow, particularly under different storm characteristics. Hydrological models therefore complement MCDA by offering a mechanistic representation of rainfall-runoff response, allowing the derivation of runoff and streamflow indicators that can strengthen hazard/risk interpretation (Arnold *et al*., 1998; Gassman *et al*., 2007). In the context of a Lagos flood-risk portal, hydrological simulation is especially valuable for supporting scenario exploration, such as "what happens to runoff response under higher rainfall totals or different rainfall temporal patterns?", which a static MCDA map alone cannot answer.

SWAT was selected because it is one of the few freely available watershed models compatible with the types of data available for this study. SWAT is a semi-distributed watershed model that simulates hydrology, sediment, and nutrient transport using topography, land cover, soil properties, and climate data as inputs (Arnold *et al*., 1998; Gassman *et al*., 2007). It is not a perfect fit for a highly urbanised area like Lagos, and these limitations are discussed further below. SWAT subdivides a watershed into sub-basins and hydrologic response units (HRUs), enabling spatial differentiation of land use-soil-slope combinations. Although SWAT is not primarily a city-scale drainage model, it can generate indicators relevant to flood risk analysis, including surface runoff

depth, streamflow response, and sub-basin-scale runoff contributions, which can be integrated into GIS-based susceptibility/risk frameworks as proxy indicators of hydrological response.

Recent developments in SWAT+ reorganize SWAT's structure to better represent landscape-channel connectivity and routing, with the objective of improving process representation and flexibility in complex watershed systems (Bieger *et al*., 2017). This is relevant for coastal and peri-urban systems where connectivity between uplands, channels, wetlands, and urbanized areas shapes flood propagation. However, even with SWAT+, there are notable limitations for Lagos-style flooding: SWAT/SWAT+ may not explicitly represent engineered stormwater networks and tidal backwater effects, which can be central in low-lying coastal megacities (Nkwunonwo *et al*., 2016). For this reason, the literature supports a pragmatic integration: SWAT outputs are used as hydrologically meaningful indicators of runoff generation and flow response, while GIS-based susceptibility mapping captures fine-scale conditioning factors and local terrain constraints.

A major challenge in Nigeria is the availability of long-term discharge or water level observations for calibration and validation. Nigerian SWAT applications have responded by incorporating alternative datasets and multi-site strategies to improve credibility in data-scarce conditions. Odusanya *et al*. (2019), for example, demonstrated the use of satellite-based evapotranspiration for multi-site calibration and validation of SWAT in a southwestern Nigerian catchment, illustrating an approach that reduces dependence on dense gauge networks. Such strategies align with the constraints expected for Lagos-related modelling, where gauging data may be limited, fragmented, or difficult to access consistently. For a web portal that includes a scenario runner, the goal is therefore not to claim perfect flood forecasting but to support comparative scenario reasoning (e.g., relative change in runoff/streamflow indicators under different rainfall inputs), alongside transparent reporting of uncertainty and model limitations (Odusanya *et al*., 2019; Nwogu *et al*., 2025).

The key takeaway from the SWAT/SWAT+ literature is that hydrological simulation adds value when positioned appropriately: not as a replacement for susceptibility mapping in complex urban flood contexts, but as a systematic source of runoff and streamflow response indicators that strengthen the physical basis of risk layers and enable scenario-driven planning.

2.4 Data Foundations for Flood Portals and Repeatable Workflows

Repeatability is considered central to the credibility and operational usefulness of flood risk mapping, particularly in a dynamic setting like Lagos where land cover and exposure patterns change rapidly. The project's emphasis on a portal further raises the standard: datasets must be not only scientifically defensible but also sufficiently consistent and accessible to support updates and transparent explanation.

For rainfall forcing and monitoring, CHIRPS provides long-term gridded precipitation estimates by combining satellite observations with station data, widely used for rainfall monitoring, extremes characterization, and hydrological modelling support in data-sparse regions (Funk *et al*., 2015). In Nigeria, national agencies such as NiM*et al*so provide seasonal and synoptic climate information that can complement gridded products. NiMet's annual Seasonal Climate Prediction (SCP) highlights the institutional role of forecast and monitoring products as early warning and planning inputs (NiMet, 2024). For portal design, the key literature implication is that rainfall information should be treated as an updatable input: new rainfall accumulations and scenarios should be capable of triggering refreshed runoff indicators and revised risk summaries, rather than being embedded in static maps.

Topographic representation is typically built from SRTM DEM data, which underpins terrain derivatives such as slope, flow accumulation, and drainage delineation used in susceptibility mapping and hydrological model setup (Farr *et al*., 2007). That said, the Lagos context introduces a known limitation: low-relief coastal terrain amplifies the influence of vertical error and hydrological conditioning assumptions on derived drainage patterns. The Lagos flood modelling review by

Nkwunonwo *et al*. (2016) emphasizes that modelling choices and data quality strongly influence outputs in such environments, reinforcing the need for careful DEM preprocessing and transparency in portal documentation.

To improve hydrographic consistency and basin delineation, studies commonly integrate HydroSHEDS/HydroRIVERS/HydroBASINS products derived from spaceborne elevation data, providing standardized river networks and sub-basin boundaries (Lehner *et al*., 2008). These products are frequently used to complement DEM-derived drainage layers and to support hydrological model structuring, particularly where local hydrography datasets are incomplete or inconsistent (HydroSHEDS, 2025; Lehner *et al*., 2008). In a repeatable workflow, HydroSHEDS-type products help reduce ad hoc digitization and promote consistent basin and river representations across updates.

Land cover information is essential for both susceptibility mapping (through imperviousness, wetlands, vegetation) and hydrological model parameterization. ESA WorldCover provides a 10 m global land cover product that includes built-up and wetland classes of high relevance to Lagos flood susceptibility mapping and urban expansion assessment (Zanaga *et al*., 2022). Lagos-focused land cover change work demonstrates that incorporating land cover dynamics is not optional: it is necessary for keeping flood susceptibility mapping aligned with the city's evolving surface characteristics (Idowu and Zhou, 2021). For the portal, this implies that the land cover layer should be treated as a versioned dataset, so that users can understand which year or version is being used and what implications this has for risk interpretation.

Soil properties influence infiltration capacity and runoff generation, and therefore commonly appear both in susceptibility mapping (as an infiltration/permeability proxy) and SWAT/SWAT+ parameterization. SoilGrids provides machine-learning-derived gridded soil information suitable for broad-scale modelling and comparative analysis across regions, offering a practical foundation where local soil surveys are limited (Hengl *et al*., 2017). The implication for Lagos risk mapping is

that soil proxies can enrich susceptibility modelling, but uncertainties must be acknowledged, especially where urban fill materials and engineered soils differ from natural soil profiles.

In addition to global datasets, Nigerian institutional sources provide essential contextual and operational evidence. NIHSA's Annual Flood Outlook publications represent a recurring national flood risk communication mechanism and can provide guidance on flood-prone Local Government Areas (LGAs) and seasonal concerns that can contextualize portal outputs (NIHSA, 2024). NiMet provides data request mechanisms for meteorological datasets and products that may support localized validation or bias checking where access is feasible (NiMet, n.d.). State-level materials, such as LASEMA's flood vulnerability assessment and mapping documentation, illustrate the institutional demand for mapped vulnerability and early warning support in Lagos (LASEMA, n.d.). Post-event assessments, such as IOM's Joint Post-Flood Assessment for Lagos State, provide structured event evidence that can support validation and portal narrative context (IOM, 2024).

Finally, literature on flood information systems indicates that flood risk products become more actionable when they are delivered through web GIS platforms that enable exploration, comparison, and updates. Rodrigues da Silva *et al*. (2023) described the RiverCure Portal as a Web GIS platform designed to manage flood data, support simulation workflows, and disseminate results to different stakeholders, highlighting the broader movement toward web-enabled flood decision support. Beyond system design, the literature emphasizes interoperability and accessibility through standards. The Open Geospatial Consortium (OGC) Web Map Service (WMS) standard provides a widely used interface for requesting and displaying georeferenced map images across clients, enabling consistent publication of map layers for decision support systems (OGC, 2006; OGC, n.d.). For a Lagos flood-risk portal, these lessons imply that the portal should not only display final risk maps but also expose supporting factor layers, enable user-friendly querying and legends, and support update workflows that preserve transparency and reproducibility.

2.5 Tabular Summary of Reviewed Literature

Table 2.1 summarizes key literature and institutional sources reviewed, emphasizing Lagos/Nigeria studies and the methods most relevant to a QGIS-based MCDA/AHP + SWAT/SWAT+ + web portal workflow.

Table 2.1: Summary of reviewed literature

Author/ Year	**Study Area**	**Data Used**	**Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)**	**Key Findings**	**Relevance to this project**
Adelekan (2010)	Lagos, Nigeria	Community/ urban evidence	Vulnerability analysis (urban flooding)	Flood impacts are uneven; poorest and coastal communities often face disproportionate vulnerability	Supports risk framing beyond hazard; informs portal communication of vulnerability
Ajibade *et al*. (2013)	Lagos, Nigeria	Household/ community evidence	Vulnerability and resilience analysis	Gendered and socio-economic patterns	Reinforces need to represent risk, not only

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
				influence flood vulnerability and recovery	hazard, in portal outputs
Nkwunonwo *et al*. (2016)	Lagos region, Nigeria	Review of models and data issues	Literature synthesis of flood modelling	Highlights modelling/data limitations in Lagos and challenges for urban flood risk management	Justifies transparent, well-documented workflows and uncertainty reporting
Adelekan (2016)	Lagos, Nigeria	Policy/urban context	Flood risk management synthesis	Flood risk management in Lagos is shaped by governance, exposure, and planning	Motivates portal as a planning-support and risk communicati

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
				constraints	on tool
Adelekan and Asiyanbi (2016)	Lagos, Nigeria	Community survey evidence	Risk perception analysis	Risk perception influences preparedness and coping, implying communication needs	Supports emphasis on interpretability and usability of portal maps
LASEMA (n.d.)	Lagos State	State mapping materials	Institutional vulnerability/risk mapping	Institutional recognition that mapping supports early warning and response	Provides local precedent and evidence of stakeholder need
Idowu and Zhou (2021)	Lagos State	LULC + flood hazard context	GIS hazard mapping with LULC change	Land cover change is central to	Supports inclusion of updatable

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
			emphasis	flood hazard dynamics in Lagos	land cover layers and built-up/wetl and factors
Isiaka *et al*. (2023)	Lagos State	Terrain/land cover factors	Frequency ratio susceptibility mapping	Demonstrates data-driven susceptibility mapping for Lagos using conditioning factors	Provides comparative evidence and validation expectations for Lagos susceptibility mapping
Adeaga *et al*. (2020)	Lagos metropolis (part)	GIS layers (multi-factor)	Flood vulnerability mapping	Demonstrates GIS-based vulnerability zoning for Lagos sub-areas	Supports factor selection trends and local validation

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
					approaches
Ajibade *et al*. (2021)	Ibadan, Nigeria	GIS conditioning factors	AHP-based flood-prone mapping	AHP provides structured weighting for flood-prone mapping	Supports MCDA/AHP methodological foundation under Nigerian conditions
Nkeki *et al*. (2022)	Ona River Basin, Nigeria	Raster factors, limited observations	Analytic hierarchical raster fusion (AHP-informed)	Demonstrates feasible flood risk mapping under data scarcity	Reinforces MCDA/AHP feasibility and limitations relevant to Nigeria
Odusanya *et al*. (2019)	Southwestern Nigeria (catchment)	SWAT inputs + satellite ET	SWAT calibration/validation in data-scarce context	Satellite-based ET supports calibration where	Supports SWAT integration and scenario

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
				discharge data are sparse	reasoning under limited gauges
Arnold *et al*. (1998)	Global (model development)	Model framework	SWAT development	Introduces SWAT structure and watershed-based modelling logic	Core foundation for SWAT-based runoff indicators used in risk analysis
Gassman *et al*. (2007)	Global (review)	Multiple case studies	SWAT review	Summarizes SWAT applications and research directions	Supports appropriate positioning of SWAT in catchment-scale risk support
Bieger *et al*. (2017)	Global (model	Model	SWAT+	SWAT+ restructures	Supports discussion of

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
	evolution)	framework	introduction	SWAT to improve connectivity/routing representation	SWAT+ suitability and advances for scenario support
Funk *et al*. (2015)	Global/ Africa coverage	CHIRPS rainfall	Dataset development	CHIRPS supports long-term rainfall monitoring and extremes analysis	Justifies CHIRPS as repeatable rainfall forcing for portal scenario updates
Farr *et al*. (2007)	Global	SRTM DEM	Dataset development	SRTM provides global DEM foundation for terrain derivatives	Supports DEM-based conditioning factors and SWAT watershed

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
					setup
Lehner *et al*. (2008)	Global	HydroSHEDS	Dataset development	HydroSHEDS provides standardized hydrography derived from elevation data	Supports repeatable river/basin representation across updates
Hengl *et al*. (2017)	Global	SoilGrids	Dataset development	SoilGrids provides global soil properties via ML	Supports soil infiltration proxy and SWAT parameterization inputs
Zanaga *et al*. (2022)	Global	ESA WorldCover	Dataset development	10 m land cover supports built-up/wetland identification	Supports land cover factor layers and exposure proxies
NIHSA	Nigeria	Flood outlook	Institutional risk	Annual flood	Provides

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
(2024)	(national)	publications	communication	outlook supports seasonal preparedness and risk communication	national context and evidence stream for portal narrative/updates
NiMet (2024)	Nigeria (national)	Seasonal Climate Prediction	Institutional climate outlook	Seasonal climate information supports early warning and planning	Supports rainfall-context communication and scenario framing in portal
Rodrigues da Silva *et al*. (2023)	Portugal case (generalizable)	Flood data + modelling outputs	Web GIS portal design + simulation support	Web GIS platforms can support flood simulation	Provides a literature basis for portal

Author/ Year	Study Area	Data Used	Method (MCDA/AHP, SWAT/SWAT+, remote sensing, portal approach, validation)	Key Findings	Relevance to this project
				workflows and dissemination	features: layers, simulation outputs, stakeholder access
OGC (2006)	Global standard	WMS specification	Web map standard	WMS enables interoperable serving of map layers to web clients	Supports portal design choices for interoperable map dissemination

2.6 Knowledge Gaps

Despite all the research and growing institutional mapping activity, several gaps were identified that directly motivated the development of a Lagos-focused flood risk analysis and visualization web portal:

Limited evidence of integrated, user-friendly flood risk portals focused on Lagos for non-specialist use. Many Lagos studies produce valuable susceptibility/vulnerability maps, yet dissemination is often confined to static figures in papers or reports, limiting interactive exploration by planners and communities (Nkwunonwo *et al*., 2016; Nwogu *et al*., 2025; LASEMA, n.d.).

Many susceptibility/risk mapping outputs are "one-off" products rather than repeatable pipelines that can be updated with new rainfall or land cover information. Lagos is rapidly changing, and the literature emphasizes the importance of keeping land cover representations current and documenting modelling assumptions for reuse (Idowu and Zhou, 2021; Nkwunonwo *et al*., 2016).

Weak integration between hydrological simulation indicators and transparent MCDA/AHP susceptibility mapping in Lagos-oriented workflows. Some studies emphasize GIS factor overlays, while others focus on hydrological modelling in data-scarce basins; fewer studies demonstrate an operational integration where runoff/streamflow indicators systematically strengthen susceptibility/risk layers for decision support (Odusanya *et al*., 2019; Nkeki *et al*., 2022; Idowu and Zhou, 2021).

Fragmented flood evidence and validation inventories limit confidence and comparability across studies. Post-event assessments and agency reports exist, but structured flood inventories and standardized validation practices remain uneven, constraining reproducibility and uncertainty interpretation (IOM, 2024; Nkwunonwo *et al*., 2016; NIHSA, 2024).

Uncertainty reporting and communication remain insufficiently operational for planning. Reviews of Nigerian flood risk mapping highlight the broader challenge of translating technical outputs into planning instruments and governance processes, indicating a need for tools that embed transparency, documentation, and user-facing explanations (Nwogu *et al*., 2025; Adelekan, 2016).

2.7 Summary

The literature establishes that flooding in Lagos comes from a tangled interaction of low-lying coastal terrain, intense rainfall and stormwater build-up, drainage problems and poor maintenance, and rapid land cover change, all of it shaped by unequal exposure and vulnerability (Adelekan, 2010; Ajibade *et al*., 2013; Idowu and Zhou, 2021). GIS-based flood susceptibility mapping remains widely used in Nigeria because it is feasible under data limitations and can integrate multiple conditioning factors, especially when implemented through transparent MCDA/AHP methods that support interpretability and stakeholder communication (Saaty, 1987; Ajibade *et al*., 2021; Nkeki *et al*., 2022). However, the Lagos literature also shows that susceptibility mapping alone may miss dynamic rainfall-runoff behavior, motivating the inclusion of hydrological modelling, particularly SWAT/SWAT+, to generate runoff and streamflow response indicators that strengthen hazard and scenario interpretation (Arnold *et al*., 1998; Bieger *et al*., 2017; Odusanya *et al*., 2019).

The review also demonstrated that producing maps alone is not sufficient: risk information must be accessible, updatable, and understandable to diverse users. Web GIS platforms and interoperability standards such as OGC WMS support these goals by enabling interactive layer publication, dashboards, and scenario-driven exploration (Rodrigues da Silva *et al*., 2023; OGC, 2006). Consequently, the reviewed evidence justifies a workflow in which QGIS-based spatial modelling, SWAT/SWAT+ hydrological indicators, and MCDA/AHP risk mapping are operationalized within a web portal that supports transparent risk communication, repeatable updates, and scenario-based decision support. These insights provide the conceptual and technical basis for the methodology presented in Chapter Three.

CHAPTER THREE

METHODOLOGY

This chapter presents the methodology employed to build the flood risk portal, covering data collection and cleaning, hydrological modelling, flood map creation, and web application development. Each step has been documented with sufficient detail to enable reproducibility. The methodology combines Geographic Information System (GIS)-based spatial analysis, Soil and Water Assessment Tool (SWAT) hydrological modelling, multi-criteria flood susceptibility mapping, and web development. It draws on established flood risk assessment and hydrological modelling methods, adapted for the specific conditions in Lagos State.

3.1 Research Design and Approach

3.1.1 Overall Research Paradigm

The approach is quantitative and GIS-based, bringing together spatial data analysis and hydrological simulation. Philosophically, it sits within a positivist framework, relying on measurable, empirical datasets and reproducible computational methods to assess flood risk across Lagos State. This paradigm is well-established in flood hazard research in Nigeria (Adeaga *et al*., 2020) and internationally (Tehrany *et al*., 2015). A quantitative approach makes sense here because flood risk is fundamentally spatial: it can be characterized through terrain attributes, proximity to water bodies, rainfall intensity, land cover patterns, and hydrological response—all of which are measurable and amenable to GIS analysis (Malczewski, 1999).

3.1.2 Project Workflow Overview

The project follows a reproducible pipeline with four linked components, each feeding its outputs into the next:

- Component 1 – Data Acquisition and Preprocessing: This component forms the foundation of the workflow. Rainfall, terrain, river network, land cover, and soil data were assembled from various open sources and harmonised to a common coordinate system and resolution. This proved to be one of the most labour-intensive parts of the project. All datasets will be harmonized to a common coordinate reference system (EPSG:32631, UTM Zone 31N), clipped to the Lagos State boundary, resampled to consistent resolutions, and subjected to quality control checks.
- Component 2 – Hydrological Modelling with SWAT+: Using the preprocessed terrain, land cover, soil, and climate data, the SWAT+ model was set up for a representative catchment draining into the Lagos Lagoon system. The model was calibrated and validated against a monthly discharge reference (GEOGLOWS v2 retrospective discharge, adopted after gauged NIHSA records could not be obtained), and scenario simulations generate runoff and peak-flow indicators under baseline and extreme rainfall conditions (Arnold *et al*., 1998; Bieger *et al*., 2017).
- Component 3 – Flood Susceptibility and Risk Mapping: Terrain-derived indices (elevation, TWI, slope, flow accumulation), proximity to waterways, land cover, extreme rainfall, and an SCS Curve Number runoff estimate were integrated into a multi-criteria flood susceptibility index using the Analytical Hierarchy Process (AHP) for weighting (Saaty, 1980). Susceptibility maps were combined with exposure data to produce flood risk maps.
- Component 4 – Web Portal Development and Deployment: An interactive web portal was designed and built as a single-process Streamlit application with a Folium/Leaflet map to publish map layers, time-series dashboards, and scenario exploration tools for planners and decision-makers. A heavier spatial-database and tile-server stack was considered but not adopted, as it offered no analytical benefit for the static raster and tabular outputs of this study (Section 3.7).

3.1.3 Justification for Methodological Choices

Several reasons underpin the choice of GIS-based multi-criteria analysis coupled with SWAT hydrological modelling. First, Lagos State's flood risk is driven by a complex interplay of physical and anthropogenic factors—low-lying coastal terrain, extensive lagoon systems, rapid urbanization, inadequate drainage infrastructure, and intense tropical rainfall—that require an integrated spatial framework for assessment (Adelekan, 2010; Obiefuna *et al*., 2021). Second, the SWAT model has been widely and successfully applied in tropical West African catchments and provides physically based estimates of runoff response to rainfall, land use, and soil conditions (Schuol *et al*., 2008; Awotwi *et al*., 2015). Third, the AHP-based multi-criteria approach offers transparency and interpretability, enabling stakeholders to understand and interrogate the weighting of different flood factors (Saaty, 1980; Malczewski, 1999). Fourth, a web portal addresses the critical gap identified by Adeaga *et al*. (2020) and LASEMA (n.d.) whereby flood risk information in Lagos remains fragmented, inaccessible to non-specialists, and difficult to update when new data become available.

3.2 Study Area Description

3.2.1 Geographic and Administrative Setting

Lagos State is situated in the southwestern corner of Nigeria, bounded by longitudes 2°42′E to 3°42′E and latitudes 6°22′N to 6°52′N. It covers roughly 3,577 km² in total, with about 22% of that made up of lagoons, creeks, and wetlands (Obiefuna *et al*., 2021). Even though it is the smallest state in Nigeria by land area, Lagos is the most populous, home to an estimated 20 million-plus residents (National Bureau of Statistics, 2022). The state is divided into 20 Local Government Areas (LGAs) and is bounded by Ogun State to the north and east, the Atlantic Ocean to the south, and the Republic of Benin to the west. The state's administrative and economic importance as Nigeria's commercial hub makes flood risk management a matter of national significance (Adelekan, 2010).

3.2.2 Climatic Characteristics

Lagos State experiences a tropical wet and dry climate (Köppen classification: Aw) characterized by two distinct seasons. The wet season runs from April to October, with the heaviest rains falling in June and September, separated by a short dry spell (the “August Break”) in July–August. The dry season lasts from November to March, with very little rainfall. Average annual rainfall sits between roughly 1,500 mm along the coast and 1,800 mm in parts of the mainland (Adelekan, 2010; Nigerian Meteorological Agency [NiMet], 2023). Downpours can be extreme, with daily totals sometimes exceeding 150 mm during severe events, enough to overwhelm the limited drainage network and set off widespread surface and river flooding (Ajibade *et al*., 2013). Average temperatures range from 25°C to 29°C through the year, and relative humidity stays around 70–80%.

3.2.3 Hydrological and Terrain Characteristics

Lagos State is characterized by extremely low relief, with most of the land surface lying below 20 m above mean sea level, and extensive areas below 5 m (Adeaga *et al*., 2020). The terrain is dominated by a barrier–lagoon coastal system comprising the Lagos Lagoon, Lekki Lagoon, Ologe Lagoon, and interconnected creeks and channels. Major rivers draining into the lagoon system include the Ogun River (the largest), Aye River, Majidun River, and Owo River. The flat terrain, high water table, and tidal influence on the lagoon system all mean that natural drainage is slow to begin with, and every new stretch of impervious surface from urban development only adds more runoff while cutting the ground's ability to absorb water (Obiefuna *et al*., 2021). All of this leaves Lagos exposed to flooding from several directions at once: direct rainfall pooling on hard surfaces, rivers breaking their banks, lagoon surges driven by tides, and events where two or more of these happen together.

3.2.4 Urbanization and Flood Vulnerability Context

Lagos has experienced rapid and largely unplanned urbanization, with built-up area expanding by over 400% between 1984 and 2020 (Braimoh and Onishi, 2007; Obiefuna *et al*., 2021). Much of what is now dense residential housing in areas like Ajah, Ibeju-Lekki, and parts of Ikorodu was farmland or marshland within living memory. This pattern of development has drastically reduced the land's ability to absorb rainwater. The expansion has led to encroachment on floodplains, wetlands, and drainage channels. Informal settlements, particularly in low-lying areas adjacent to lagoons and waterways (e.g., Makoko, Badia, Ajegunle), are the most severely affected by flood events (Ajibade *et al*., 2013). The Lagos State Emergency Management Agency (LASEMA) regularly responds to flood emergencies during the wet season, and significant flooding events were recorded in 2010, 2011, 2012, 2015, 2017, 2020, and 2021 (LASEMA, n.d.; Adelekan, 2016). These conditions provide strong justification for an integrated flood risk analysis and visualization system.

3.2.5 Specific Focus Catchments for SWAT Modelling

While the web portal will support statewide viewing and querying, the SWAT hydrological modelling component will focus on one or more representative catchments draining into the Lagos Lagoon system. The primary candidate is the Ogun River catchment at its lower reach within Lagos State, which is the largest contributor of fluvial discharge to the Lagos Lagoon and was responsible for catastrophic flooding events in 2010, 2011, and 2012 following releases from Oyan Dam upstream (Adelekan, 2016). A secondary candidate is the Aye–Majidun River catchment in the Ikorodu LGA area, which experiences recurrent flooding. This selection ensures that SWAT modelling captures the dominant fluvial flood processes affecting Lagos while remaining computationally feasible within the scope of a Master’s thesis. The exact catchment boundaries will be delineated from the conditioned Digital Elevation Model (DEM) during model setup (Section 3.5.1).

3.3 Data Collection and Sources

The following subsections describe each dataset employed in the study, its source, and its intended use. The study relied primarily on free, globally available datasets, supplemented by local agency data where I could get it, following the approach recommended for flood risk studies in data-sparse African environments (Schuol *et al*., 2008; Funk *et al*., 2015).

3.3.1 Rainfall Data

Rainfall is the single biggest driver of flooding in Lagos and was therefore the first dataset targeted for acquisition. Spatially continuous gridded data were needed to cover the entire state, along with ground-based gauge records for validation of the satellite estimates. the study employed two complementary rainfall data sources:

(a) CHIRPS Gridded Rainfall Data

The Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) is a quasi-global rainfall dataset that merges satellite cold-cloud-duration estimates with in-situ gauge observations to produce daily and pentadal (5-day) rainfall grids at 0.05° (≈5.3 km) spatial resolution (Funk *et al*., 2015). CHIRPS covers the period from 1 January 1981 to near-present and is freely available from the Climate Hazards Center at the University of California, Santa Barbara.

Specifications: Spatial resolution: 0.05° × 0.05° (~5.3 km at the equator). Temporal resolution: daily (CHIRPS daily v2.0). Temporal coverage: 1 January 1981 to 31 December 2023 (the most recent complete calendar year available at the time of data acquisition). Geographic extent for download: latitude 6.2°N to 7.0°N, longitude 2.5°E to 4.0°E (buffered beyond Lagos State boundary to ensure complete coverage). Data format: GeoTIFF rasters (one file per day or monthly composite). Coordinate reference system: WGS 84 (EPSG:4326).

Access procedure: CHIRPS data will be downloaded from the official CHIRPS data portal (Initial attempts at bulk download via FTP encountered connection timeouts, necessitating a switch to

downloading individual monthly files via HTTP). The data will be obtained from the CHIRPS data portal at https://data.chc.ucsb.edu/products/CHIRPS-2.0/ using automated Python scripts with the 'requests' or 'wget' library. For daily data, navigation was performed to the '/daily/' subdirectory and download files for the period 1981–2023. Alternatively, CHIRPS data can be accessed through Google Earth Engine (GEE) using the dataset identifier 'UCSB-CHG/CHIRPS/DAILY'.

Intended use: (i) Computation of rainfall indices for flood susceptibility mapping, including annual maxima, seasonal totals, and extreme percentiles (e.g., 95th and 99th percentile daily rainfall). (ii) Preparation of precipitation forcing time series for the SWAT hydrological model. (iii) Generation of rainfall scenario inputs, including extreme event extraction and rainfall multiplier scenarios for sensitivity analysis. (iv) Time-series visualization in the web portal dashboards.

Quality control: the downloaded data were inspected the downloaded CHIRPS time series for gaps, anomalies, and missing values. Specifically, the following steps were applied: (a) checking for missing days in the daily time series and fill gaps of up to 3 consecutive days using linear interpolation; (b) flag and investigate any daily values exceeding 300 mm as potential artefacts; (c) compare CHIRPS totals against the NiMet Ikeja gauge record over the overlapping period to assess bias magnitude, treating the NiMet gauge as the ground-based reference; and (d) compute correlation coefficients and percent bias between CHIRPS and gauge data at monthly time steps to quantify agreement.

(b) NiMet Gauge Records (Ikeja Station)

Ground-based meteorological records were obtained from the Nigerian Meteorological Agency (NiMet) for the Ikeja synoptic station (Murtala Muhammed International Airport; approximately 6.58 degrees North, 3.33 degrees East, 38 m above sea level), which is the principal long-record station for the Lagos metropolitan area (Section 1.2). The record is a continuous 30-year annual series covering 1996 to 2025, comprising annual rainfall, temperature and relative humidity, with no missing years.

Intended use. The NiMet Ikeja record serves two purposes in this study. First, it is the ground-based reference against which the CHIRPS satellite rainfall estimates are validated: CHIRPS is cross-checked against the Ikeja gauge over the overlapping period, giving a correlation of r = 0.77 and a mean bias of approximately -95 mm. NiMet is therefore the reference and CHIRPS the product being assessed, and not the other way round. Second, the 30-year series is the basis for the design-rainfall frequency analysis reported in Chapter Four (Section 4.7.3): the Gumbel extreme-value distribution and the associated return-period rainfall estimates were fitted directly to the NiMet Ikeja annual-maximum series, which supplies the multi-decadal record length that a robust extreme-value fit requires. The trend tests reported in Chapter Four (Mann-Kendall on annual rainfall and on mean annual temperature) were likewise computed on this series.

Quality control. The NiMet annual series was screened for missing years, for outliers exceeding three standard deviations from the long-term mean, and for step changes in the annual totals that might indicate a station relocation or an instrument change; no gaps were present across the 30-year record. The screened series was then used both as the CHIRPS validation reference and as the input to the frequency analysis.

3.3.2 Terrain and Digital Elevation Model (DEM) Data

Terrain data underpin both flood susceptibility analysis and hydrological modelling because they control flow paths, accumulation patterns, slope gradients, and where watershed boundaries fall. The study employed the Copernicus GLO-30 Global Digital Elevation Model (DEM) as its terrain source.

Dataset: Copernicus GLO-30 DEM, derived from the TanDEM-X mission and distributed by the European Space Agency. Spatial resolution: 1 arc-second (approximately 30 m at the equator). Coverage: near-global. Data format: GeoTIFF. Coordinate reference system: WGS 84 (EPSG:4326), with elevation in metres. The product is openly available without registration.

Access procedure: Six GLO-30 tiles covering the full Ogun-Lagos basin were downloaded from the Amazon Web Services open-data bucket `copernicus-dem-30m` and mosaicked using GDAL's `gdal_merge.py` utility to produce a seamless DEM for the study area.

Justification for the change from SRTM. The original methodology specified the Shuttle Radar Topography Mission (SRTM) DEM. During data assembly, the available SRTM tiles were found to terminate at 7 degrees North and therefore did not cover the northern headwaters of the Ogun basin, which extend to approximately 8.95 degrees North and drain into the Lagos Lagoon system. Because the hydrological model requires the complete contributing catchment, SRTM could not be used for basin delineation. Copernicus GLO-30 provides equivalent 30 m resolution with complete basin coverage, improved vertical accuracy relative to the earlier SRTM product, and open access without registration, and was therefore adopted in its place. Higher-resolution LiDAR DEMs at 1-5 m, which would be preferable for the low-relief Lagos terrain, were either not publicly available or too costly to license for this project.

Resampling for delineation. The 30 m DEM comprises approximately 81 million cells over the catchment, which stalled the D8 flow-direction routine during preliminary delineation. The DEM was therefore resampled to 90 m for the catchment-delineation stage, which is standard practice for a catchment of roughly 20,000 km² and produced a stable delineation without material loss of hydrological detail. The 30 m product was retained for the terrain-derivative layers used in susceptibility mapping. The consequences of this change for the DEM-conditioning steps are noted in Section 3.4.2.

3.3.3 Hydrography and River Network Data

Accurate representation of the river and waterway network is important for computing distance-to-river metrics in flood susceptibility mapping and for checking DEM-derived stream networks against reality. the study employed the HydroSHEDS suite of products, specifically HydroRIVERS and HydroBASINS (Lehner *et al.*, 2008; Lehner and Grill, 2013).

HydroRIVERS: A global vector dataset of river centrelines derived from the HydroSHEDS framework at 15 arc-second resolution. Each river segment carries attributes including upstream drainage area, average long-term discharge estimate, stream order (Strahler), and river length. The Africa regional subset will be downloaded from https://www.hydrosheds.org/products/hydrorivers and clipped to the Lagos State extent.

HydroBASINS: A global dataset of sub-basin polygons delineated at multiple Pfafstetter coding levels. the study employed Level 08 and Level 12 basin delineations for Africa to define hydrological units for aggregation and reporting within the portal. Available from https://www.hydrosheds.org/products/hydrobasins.

Intended use: (i) Computation of Euclidean distance-to-nearest-river raster for flood susceptibility analysis. (ii) Computation of river network density (km of river per km²) within grid cells or sub-basins. (iii) Consistency checking against DEM-derived stream networks during SWAT setup. (iv) Reference hydrography layer for the web portal map display.

3.3.4 Land Cover Data

Dataset: ESA WorldCover 2021, version 2.0. Spatial resolution: 10 m. Classification: 11 global land cover classes based on Sentinel-1 and Sentinel-2 imagery. Data format: GeoTIFF. Coordinate reference system: WGS 84 (EPSG:4326) (Zanaga *et al*., 2022).

Access procedure: WorldCover tiles covering Lagos State will be downloaded from the ESA WorldCover portal at https://esa-worldcover.org/en or via the Terrascope platform. The relevant tiles will be mosaicked and clipped to the Lagos State boundary.

Intended use: (i) Classification of land cover into SWAT-compatible land use categories for Hydrologic Response Unit (HRU) definition. (ii) Identification of built-up areas and impervious surfaces as a key flood susceptibility factor. (iii) Detection of wetland and open water areas. (iv) Land cover layer for the web portal. The 10 m resolution of ESA WorldCover provides

substantially better spatial detail than older products (e.g., GlobCover at 300 m) and is essential for capturing Lagos's heterogeneous urban landscape (Zanaga *et al*., 2022).

3.3.5 Soil Data

Dataset: ISRIC SoilGrids 2.0 (Hengl *et al*., 2017; Poggio *et al*., 2021). Spatial resolution: 250 m. Variables: sand content (%), silt content (%), clay content (%), organic carbon content (g/kg), bulk density (kg/m³), and derived hydrological properties including saturated hydraulic conductivity. Depth intervals: 0–5 cm, 5–15 cm, 15–30 cm, 30–60 cm, 60–100 cm, and 100–200 cm.

Access procedure: SoilGrids data will be accessed through the ISRIC WCS (Web Coverage Service) at https://soilgrids.org/ or via direct download from https://files.isric.org/soilgrids/latest/data/. data were downloaded the sand, silt, clay, organic carbon, and bulk density layers for the 0–30 cm depth interval (representing the root zone and surface runoff generation layer) and clip to the study area.

Intended use: (i) Derivation of soil hydrological group assignments (A, B, C, D based on infiltration characteristics) for the SWAT model and the SCS Curve Number method. (ii) Computation of a soil infiltration capacity proxy raster for flood susceptibility mapping. (iii) Preparation of SWAT-compatible soil parameter tables including available water capacity, saturated conductivity, and moist bulk density for each soil layer.

3.3.6 Discharge Data for Model Calibration

Observed river discharge records are required for SWAT+ model calibration and validation. Discharge data availability for Nigerian rivers is historically limited owing to sparse gauging-network coverage and gaps in record-keeping (Awotwi *et al*., 2015).

Source originally sought. The Nigeria Hydrological Services Agency (NIHSA), which maintains stream-gauge records for the Ogun River and other monitored watercourses, was approached with a formal request for monthly and, where available, daily discharge records for gauging stations on the

Ogun River within or upstream of Lagos State. NIHSA discharge data were not obtained within the project period. The request is recorded here because the resulting reliance on a modelled discharge reference is a material feature of the study that is revisited in the limitations (Section 1.5) and in Chapter Four.

Calibration reference actually used. In place of gauged records, the model was calibrated against GEOGLOWS version 2 ECMWF retrospective discharge at reach 140585128, located at Kara/Akute on the Ogun main stem above the Lagos Lagoon. This series comprises 31,589 daily values covering 1940 to 2026, aggregated to monthly means for model evaluation. GEOGLOWS is itself a modelled product, so the performance statistics reported in Chapter Four measure agreement between two models rather than agreement with observation; this is stated plainly because it bears directly on the strength of the calibration claim. Calibrating against a continental-scale modelled discharge reference in the absence of local gauging follows the data-scarce precedent set by Odusanya *et al*. (2019) for southwestern Nigerian catchments, and its principal weakness is partly offset by the independent evapotranspiration validation introduced in Section 3.8.1 and reported in Chapter Four.

3.3.7 Flood Inventory and Exposure Data

Spatial flood inventory data are required for validating the flood susceptibility map by comparing predicted high-susceptibility zones against observed flooding. Because no comprehensive spatial flood database exists for Lagos State, two independent datasets were compiled from open archives.

Flood inventory datasets. (i) A MODIS 250 m flood-extent product for Nigeria covering July to November 2012 (NASA / Dartmouth Flood Observatory, obtained through the Humanitarian Data Exchange), which provides 35.2 km² of detected flood extent within Lagos State distributed across 290 discrete patches. Permanent water bodies are masked in the source product, so the extent represents detected flooding rather than the lagoon system. (ii) NEMA 2022 flood records (obtained

through the Humanitarian Data Exchange), comprising 62 community-level reports across 13 of the 20 Lagos Local Government Areas.

No geocoding of the NEMA records was attempted. The NEMA records carry community names but no coordinates. Assigning coordinates from place names would inject fabricated locations into the dataset used for validation, and the records were therefore used only at the Local Government Area level of aggregation that the source supports. This is a deliberate methodological decision rather than an omission: it preserves the integrity of the validation by refusing to manufacture spatial precision the source does not contain.

Exposure data. Population distribution was represented using the WorldPop gridded population product for Nigeria at approximately 100 m resolution. Building-footprint and road-network layers from OpenStreetMap were incorporated where licensing and coverage permitted, with constraints documented where the data were incomplete.

Intended use. The MODIS 2012 flood extent serves as the primary independent surface for pixel-level validation of the susceptibility map (Section 3.8.2), and the NEMA 2022 records support a Local Government Area-level cross-check. Exposure layers are combined with the susceptibility classes to estimate population exposure by zone and by Local Government Area (Section 3.6.6).

3.4 Data Preprocessing and Preparation

All the datasets described in Section 3.3 need preprocessing to make them spatially consistent, clean, and compatible with downstream analysis. This section documents the exact preprocessing steps, organised by operation type. All GIS preprocessing will be performed using **QGIS 3.34 (Sketcher)**, **GDAL 3.8**, and **Python 3.11** with the **rasterio**, **geopandas**, **numpy**, and **xarray** libraries.

3.4.1 Data Harmonization (CRS, Clipping, and Resampling)

All spatial datasets will be harmonized to a common projected coordinate reference system, clipped to a consistent geographic extent, and resampled to operational resolutions as needed. The following standardized procedures will be applied:

Step 1 – Coordinate Reference System (CRS) Reprojection: All datasets will be reprojected to EPSG:32631 (WGS 84 / UTM Zone 31N), which is the appropriate UTM zone for Lagos State. This projected CRS enables distance and area calculations in metres, which is essential for terrain analysis, buffer operations, and hydrological modelling. Reprojection will be performed using GDAL's 'gdalwarp' command with bilinear resampling for continuous rasters (DEM, rainfall, soil) and nearest-neighbour resampling for categorical rasters (land cover):

gdalwarp -t_srs EPSG:32631 -r bilinear input.tif output_utm.tif

Step 2 – Clipping to Study Area: All datasets will be clipped to the Lagos State administrative boundary polygon, buffered by 5 km to avoid edge effects in terrain analysis and spatial filtering. The boundary shapefile will be obtained from the Database of Global Administrative Areas (GADM) version 4.1 (https://gadm.org/). Clipping will use GDAL's 'gdalwarp' with the '-cutline' option for rasters and 'ogr2ogr' with the '-clipsrc' option for vector datasets.

Step 3 – Resampling: Where datasets have different native resolutions, resampling will be applied to match the target analysis resolution. The target resolution for flood susceptibility analysis is 30 m, matching the Copernicus GLO-30 DEM. The ESA WorldCover (10 m) will be used at native resolution where possible but aggregated to 30 m using majority resampling for integration with other layers. SoilGrids (250 m) will be resampled to 30 m using bilinear interpolation, noting that this does not add real spatial information but ensures pixel alignment for raster overlay operations. CHIRPS rainfall (~5.3 km) will be used at native resolution for rainfall index calculation, then interpolated to 30 m only for final susceptibility map overlay.

3.4.2 DEM Conditioning and Hydrological Correction

DEM conditioning proved to be one of the more time-consuming preprocessing steps. The Copernicus GLO-30 DEM, while generally reliable, contains artefacts such as spurious sinks (depressions), vegetation canopy effects, and noise in low-relief areas that can distort derived flow paths and watershed boundaries. These issues are especially problematic in Lagos State, where the terrain is extremely flat, with most elevations below 20 m, and even small elevation errors can redirect computed flow paths (Adeaga *et al.*, 2020). the following approach was applied the following conditioning steps:

Step 1 – Void Filling: The Copernicus GLO-30 DEM may contain void (no-data) pixels in areas where the radar signal was unreliable. checks were performed for voids and fill any remaining gaps using GDAL's 'gdal_fillnodata.py' utility with a maximum search distance of 100 pixels.

Step 2 – Sink Filling: Spurious depressions (sinks) in the DEM will be filled to ensure continuous hydrological flow paths. the study employed the Wang and Liu (2006) sink-filling algorithm as implemented in SAGA GIS (accessible through QGIS Processing Toolbox) or WhiteboxTools. The algorithm identifies cells with no downslope neighbour and raises their elevation to the lowest pour point of the surrounding rim. For Lagos's flat terrain, the following was set a minimum slope gradient of 0.01% to enforce drainage directionality across flat areas. The maximum fill depth will be set to 10 m to avoid filling genuine topographic depressions such as lagoons and natural ponds.

Step 3 – Stream Burning (optional): Where DEM-derived stream networks diverge significantly from the known HydroRIVERS network, the following approach was applied stream burning (also called DEM reconditioning) by lowering DEM elevations along known river centrelines by a fixed increment (e.g., 5 m). This forces the flow direction algorithm to route water along established waterways, which is particularly important in Lagos's flat coastal environment where DEM errors easily redirect flow paths. Stream burning will use the AGREE algorithm (Hellweger, 1997) or QGIS/SAGA's 'Burn Stream Network into DEM' tool.

Step 4 – Flow Direction Computation: the following was computed flow direction from the conditioned DEM using the D8 (deterministic eight-neighbour) algorithm (O'Callaghan and Mark, 1984), which assigns each cell a single flow direction toward the steepest downslope neighbour. This algorithm is the standard for SWAT watershed delineation. Flow direction will be computed using SAGA GIS or WhiteboxTools.

Quality checks: After conditioning, the following checks were performed: (a) comparing DEM-derived stream networks at multiple accumulation thresholds (500, 1000, 5000 cells) with HydroRIVERS to visually assess agreement; (b) verify that the major rivers (Ogun, Aye, Majidun) are correctly represented; (c) check that the Lagos Lagoon system is correctly identified as a receiving water body; and (d) ensure no artificial drainage patterns have been introduced by the conditioning process.

3.4.3 Terrain Layer Derivation

Several terrain-derived layers will be computed from the conditioned DEM for use as flood susceptibility factors and SWAT inputs:

(a) Slope: Slope gradient (in degrees and percent) will be computed using the Horn (1981) algorithm as implemented in GDAL ('gdaldem slope'). Slope is a critical susceptibility factor because flatter areas have lower drainage capacity and higher ponding potential. In Lagos, where most terrain has slopes below 5°, the slope layer will help distinguish slightly elevated and better-drained areas from low-lying zones.

(b) Flow Accumulation: The flow accumulation raster will be derived from the D8 flow direction grid. Each cell's value represents the number of upstream cells draining through it, serving as a proxy for contributing area and convergent flow potential. Cells with high flow accumulation are natural drainage convergence points where water concentrates during rainfall events. Flow accumulation will be computed using SAGA GIS or WhiteboxTools.

(c) Topographic Wetness Index (TWI): The TWI quantifies the tendency of a location to accumulate water based on its contributing area and local slope. It is calculated as:

TWI = ln(a / tan β) (Equation 3.1)

where:

a = specific catchment area (flow accumulation × cell area / cell width), in m^2/m

β = local slope angle in radians

ln = natural logarithm

Higher TWI values indicate greater potential for water accumulation. This index is particularly informative in Lagos's flat terrain, where even minor topographic depressions can become sites of persistent waterlogging and flood accumulation (Beven and Kirkby, 1979). TWI will be computed using SAGA GIS.

(d) Relative Elevation (Height Above Nearest Drainage – HAND): The HAND metric represents the vertical distance of each cell above the elevation of the nearest channel cell to which it drains, computed along the flow path. HAND is a powerful flood susceptibility indicator because it directly represents the potential for inundation relative to the local drainage network (Nobre *et al.*, 2011). HAND will be computed using WhiteboxTools' 'ElevationAboveStream' tool, with the stream network defined at a flow accumulation threshold of 1000 cells (approximately 0.9 km^2 contributing area at 30 m resolution).

3.4.4 Hydrography Processing

From the HydroRIVERS vector dataset and DEM-derived stream networks, the following was computed the following spatial layers:

(a) Distance to Nearest River/Waterway: A Euclidean distance raster will be computed from the HydroRIVERS network (supplemented by Lagos Lagoon and major creek shorelines) using

GDAL's 'gdal_proximity' tool. Each cell will receive a value representing its distance (in metres) to the nearest waterway. This is one of the most important flood susceptibility factors, as proximity to water bodies is a direct determinant of fluvial and tidal flood exposure.

(b) River Network Density: the following was computed the total length of river segments (from HydroRIVERS) within a moving window (e.g., 1 km radius) or within each sub-basin polygon to create a drainage density layer. Higher drainage density indicates more extensive surface water networks and greater flood conveyance capacity, but also greater proximity exposure.

3.4.5 Rainfall Data Processing

The CHIRPS daily rainfall data for the period 1981–2023 will be processed to derive both time-series inputs for SWAT and spatial rainfall indices for susceptibility mapping: Throughout, CHIRPS provides the spatially continuous rainfall surface across Lagos State, while the NiMet Ikeja record (Section 3.3.1) is used as the ground-based reference against which the CHIRPS rainfall is validated and, where required, bias-corrected.

Step 1 – Time-series extraction: For each CHIRPS grid cell overlapping the SWAT study catchment, the following were extracted the daily rainfall time series using Python's xarray library. These time series will be formatted into the SWAT-compatible precipitation input format (station-based .pcp files with sub-basin assignments based on Thiessen polygon or nearest-cell allocation).

Step 2 – Rainfall index calculation: For flood susceptibility mapping, the rainfall conditioning factor was expressed through two monthly-aggregated spatial indices computed from the CHIRPS record: (a) mean wet-season rainfall (mm), the average rainfall accumulated over the wet-season months (April–October), representing the sustained seasonal water input that drives saturation and ponding; and (b) maximum monthly rainfall (mm), the highest monthly total recorded at each grid cell over the record, representing the intense-month condition most associated with flood generation. These indices are stored as GeoTIFF rasters at the native CHIRPS resolution and

resampled to 30 m for susceptibility overlay. The monthly framing of these indices is consistent with the seasonal terms in which flood risk is communicated operationally.

Step 3 – Scenario rainfall preparation: For SWAT scenario simulations, the following inputs were prepared modified rainfall inputs representing: (a) historical extreme events extracted from the daily record (e.g., the top 10 wettest months or storm sequences); (b) climate sensitivity scenarios applying rainfall multipliers of +10%, +20%, and +30% to the baseline daily series to simulate intensified precipitation under potential climate change conditions.

3.4.6 Land Cover Reclassification

The ESA WorldCover 2021 product classifies land cover into 11 global classes. These were reclassified into SWAT-compatible land use categories following the SWAT land use lookup table structure. The reclassification scheme is presented in Table 3.1.

Table 3.1: *ESA WorldCover to SWAT Land Use Reclassification Scheme*

		SWAT Land Use	
ESA WorldCover Class	**Code**	**Code**	**Description**
Tree cover	10	FRST	Forest (mixed)
Shrubland	20	RNGB	Range-brush
Grassland	30	RNGE	Range-grasses
Cropland	40	AGRL	Agricultural land (generic)
Built-up	50	URBN	Urban / residential
Bare/sparse vegetation	60	BARR	Barren
Snow and ice	70	N/A	Not present in Lagos
Permanent water bodies	80	WATR	Water
Herbaceous wetland	90	WETL	Wetlands

ESA WorldCover Class	Code	SWAT Land Use Code	Description
Mangroves	95	WETF	Forested wetland
Moss and lichen	100	N/A	Not present in Lagos

3.4.7 Soil Data Preparation

SoilGrids data will be processed to derive SWAT-compatible soil parameters and a flood susceptibility soil layer:

Step 1: Download sand, silt, clay, organic carbon, and bulk density rasters for the 0–5 cm, 5–15 cm, and 15–30 cm depth intervals.

Step 2: Compute depth-weighted average texture percentages for the 0–30 cm layer.

Step 3: Assign USDA soil textural classes based on the sand–silt–clay composition using the USDA soil texture triangle.

Step 4: Classify soils into hydrological soil groups (HSGs A, B, C, D) based on saturated hydraulic conductivity estimates derived from pedotransfer functions (Saxton and Rawls, 2006). Group A soils (high infiltration, sandy) will receive low susceptibility scores, while Group D soils (low infiltration, clay-dominant) will receive high susceptibility scores.

Step 5: Prepare SWAT soil parameter tables (.sol files) containing number of layers, maximum rooting depth, hydrological group, moist bulk density, available water capacity, saturated hydraulic conductivity, organic carbon content, clay/silt/sand fractions, and rock fragment content for each soil mapping unit.

3.4.8 Runoff Estimation (SCS Curve Number Method)

Surface runoff depth, which serves as factor 5 of the flood susceptibility index (Section 3.6.1), was estimated using the Soil Conservation Service Curve Number method (USDA Soil Conservation

Service, 1972, revised 1986). The method relates runoff to rainfall through a single parameter, the curve number, which encodes the combined infiltration behaviour of the soil and the overlying land cover. It was adopted because it requires only the land cover and soil layers already prepared in Sections 3.4.6 and 3.4.7, and because it is the standard approach to runoff estimation in ungauged, data-scarce settings.

A curve number was assigned to each cell from the intersection of the reclassified land cover class (Section 3.4.6, Table 3.1) and the hydrological soil group derived from the SoilGrids texture data (Section 3.4.7, Step 4), following the standard SCS lookup table. Curve numbers across Lagos State range from 55 on sandy, vegetated surfaces to 98 on open water and fully impervious surfaces.

The potential maximum retention after runoff begins, S, is obtained from the curve number as:

S = 25400 / CN – 254 (Equation 3.2)

where:

S = potential maximum soil moisture retention (mm)

CN = curve number (dimensionless, 30–100)

Equation 3.2 is the metric form of the original expression S = 1000/CN – 10, expressed in inches, multiplied by 25.4. Runoff depth is then obtained as:

$Q = (P – 0.2S)^2 / (P + 0.8S)$ for $P > 0.2S$, and $Q = 0$ otherwise (Equation 3.3)

where:

Q = runoff depth (mm)

P = rainfall depth over the accumulation period (mm)

S = potential maximum retention from Equation 3.2 (mm)

The quantity 0.2S is the initial abstraction, Ia, representing interception, surface storage and infiltration occurring before runoff begins; the ratio Ia/S = 0.2 is the standard assumption of the

original method and was retained here. Because (P + 0.8S) is equal to (P – Ia) + S, Equation 3.3 expresses runoff as the square of the effective rainfall divided by the sum of the effective rainfall and the retention capacity, so that runoff approaches zero as rainfall falls towards the initial abstraction and approaches the effective rainfall as rainfall becomes large.

The rainfall input P was the 2021 annual total from the CHIRPS record (Section 3.4.5), resampled to the 30 m analysis grid. The resulting runoff raster ranges from 0 to approximately 1,882 mm across Lagos State. Because the curve number is derived from land cover and soil rather than from a calibrated hydrological simulation, the layer is treated as a relative indicator of runoff-generating potential rather than as a predicted runoff depth; this distinction is returned to in Section 3.6.1.

3.5 SWAT/SWAT+ Hydrological Modelling Procedure

The Soil and Water Assessment Tool (SWAT) is a semi-distributed, continuous-time hydrological model developed by the USDA Agricultural Research Service (ARS) to simulate the impact of land management practices on water, sediment, and agricultural chemical yields in complex watersheds (Arnold *et al*., 1998; Neitsch *et al*., 2011). SWAT divides a watershed into sub-basins, which are further subdivided into Hydrologic Response Units (HRUs) based on unique combinations of land use, soil type, and slope class. The model simulates the hydrological cycle using a water balance approach at the HRU level:

$$SW_t = SW_0 + \Sigma(R_{day} - Q_{surf} - E_a - w_{seep} - Q_{gw}) \quad \text{(Equation 3.4)}$$

where:

SW_t = final soil water content on day t (mm)

SW_0 = initial soil water content on day 0 (mm)

R_{day} = precipitation on day i (mm)

Q_{surf} = surface runoff on day i (mm)

E_a = actual evapotranspiration on day i (mm)

w_seep = water entering the vadose zone on day i (mm)

Q_gw = return flow (groundwater contribution to streamflow) on day i (mm)

the study employed **SWAT+ (version 60.5.7 or later)** with the **QSWAT+ plugin (version 2.5 or later)** for QGIS 3.34 for model setup. SWAT+ offers improved spatial connectivity, flexible landscape routing, and enhanced reservoir and wetland representation compared to the original SWAT, making it better suited for Lagos's complex lagoon–waterway system (Bieger *et al.*, 2017). The SWAT+ executable and QSWAT+ plugin are freely available from https://swat.tamu.edu/software/plus/.

3.5.1 Model Setup and Catchment Delineation

Step 1 – DEM Input: Load the conditioned, sink-filled DEM (Section 3.4.2) into QSWAT+ as the elevation input.

Step 2 – Stream Network Definition: Define the stream network using a contributing area threshold. Given the relatively small catchment sizes in Lagos (typically 50–500 km²), the following was set the threshold to approximately 2 km² (approximately 2,222 cells at 30 m resolution). This threshold will be iteratively adjusted to produce a stream network that closely matches the HydroRIVERS reference network.

Step 3 – Outlet and Inlet Placement: the outlet was placed the catchment outlet at the point where the Ogun River enters the Lagos Lagoon (approximately 6.555°N, 3.395°E). Additional sub-basin outlets will be placed at major tributary confluences and at any locations where observed discharge data are available.

Step 4 – Watershed Delineation: QSWAT+ will delineate the watershed boundary and subdivide it into sub-basins based on the stream network and user-defined outlets. It is anticipated that the Ogun River lower catchment within Lagos State to comprise 15–30 sub-basins, depending on the threshold and outlet configuration.

3.5.2 Hydrologic Response Unit (HRU) Definition

HRUs represent unique combinations of land use, soil type, and slope class within each sub-basin. HRUs were defined HRUs using the following thresholds to filter out minor combinations and maintain computational efficiency:

Land use threshold: 10% – land use classes covering less than 10% of a sub-basin's area will be reassigned to the dominant class.

Soil threshold: 10% – soil classes covering less than 10% of a sub-basin's area will be merged with the dominant soil.

Slope threshold: 20% – slope classes covering less than 20% of a sub-basin's area will be merged. HRUs were defined three slope classes: 0–2%, 2–5%, and >5%, reflecting Lagos's predominantly flat terrain.

3.5.3 Climate Input Preparation

SWAT requires daily climate inputs including precipitation, maximum and minimum temperature, solar radiation, relative humidity, and wind speed. the following inputs were prepared these as follows:

Precipitation: CHIRPS daily rainfall, extracted for each sub-basin centroid or nearest grid cell, provides the precipitation forcing for the model, formatted as SWAT+ .pcp files. Daily CHIRPS data are retained for forcing because SWAT+ operates on a daily computational step, while model performance is evaluated at the monthly scale against the GEOGLOWS discharge reference, following the performance criteria of Section 3.5.6 (after Moriasi et al., 2007). The NiMet Ikeja record is used separately, to validate the CHIRPS rainfall and to derive the design-rainfall frequency analysis, rather than as the discharge-evaluation reference. This separation keeps the daily process representation in the model while reporting performance at the monthly scale that the reference data support.

Temperature: Daily maximum and minimum temperature data from the ERA5-Land reanalysis product (Muñoz Sabater, 2019), available at 0.1° resolution from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu/). Formatted as SWAT+ .tmp files.

Solar radiation, humidity, wind speed: If observed data are unavailable, the study employed SWAT's built-in weather generator (WGEN) populated with long-term monthly statistics derived from ERA5-Land or from the SWAT global weather database maintained at Texas A&M University (https://swat.tamu.edu/data/). The weather generator stochastically simulates daily values based on monthly means, standard deviations, and probability parameters.

3.5.4 Baseline and Scenario Simulation Design

Simulation period and warm-up. The model was run over the period 2010 to 2023, with the first year (2010) used as a warm-up during which the model initialises soil-moisture stores and groundwater aquifers (`nyskip = 1`); outputs for the warm-up year are discarded. Model evaluation therefore covers 2011 to 2023. The simulation window was set by the forcing data: the CHIRPS rainfall and ERA5-Land temperature series were extracted for 2010 to 2023, and an earlier start would have required an additional decade of forcing data that was not assembled.

Calibration period: 2011 to 2017, during which the model parameters were adjusted to match the monthly discharge reference.

Validation period: 2018 to 2023, during which the calibrated model's performance was independently assessed.

Scenario simulations. Following calibration and validation, the following scenarios were run: (a) *Baseline scenario* – the full evaluation period (2011-2023) under historical climate inputs; (b) *Extreme historical scenario* – the wettest year in the simulated record together with the 2022 flood year; and (c) *Climate-sensitivity scenarios* – uniform rainfall multipliers of +10%, +20%, and +30% applied to the baseline daily precipitation to assess runoff sensitivity. These three scenario families

satisfy the requirement of Objective (ii) for baseline, extreme-historical, and climate-sensitivity simulations.

3.5.5 Calibration and Validation Procedure

Calibration is essential, since without it the model would operate on default parameters that may bear little resemblance to conditions in the Ogun catchment. Calibration was performed through iterative manual adjustment of the SWAT+ calibration file (`calibration.cal`), with each candidate parameter set evaluated by direct execution of the SWAT+ engine (revision 62) and scored against monthly GEOGLOWS discharge at the catchment outlet using NSE, R-squared and PBIAS.

This approach was adopted in preference to the SWAT-CUP / SUFI-2 procedure originally proposed, for two reasons. First, the calibration target was a modelled rather than a gauged discharge series, for which a formal uncertainty band around the observations cannot be meaningfully interpreted. Second, the small number of sensitive parameters identified during preliminary testing did not warrant automated global optimisation. A direct consequence of not using SUFI-2 is that the P-factor and R-factor uncertainty statistics, which are outputs of the SUFI-2 95% prediction-uncertainty band, were not produced; this is recorded as a limitation in Section 1.5 and reflected in Table 3.6 and in Chapter Four.

Calibration parameters. The parameters adjusted during calibration are those of the SWAT+ revision-62 parameter database, listed in Table 3.2. The soil-evaporation compensation factor (`esco`), the plant-uptake compensation factor (`epco`), and the available water capacity (`awc`) were tested during calibration and found to be either inert or to move the simulation in the wrong direction in this SWAT+ revision; they were therefore excluded from the final parameter set.

Table 3.2: *SWAT+ calibration parameters and permitted ranges*

Parameter	Description	Permitted range
perco	Percolation coefficient	0.00 - 1.00
petco	Potential-evapotranspiration multiplier	0.80 - 1.20
canmx	Maximum canopy storage (mm)	0.00 - 100.0
alpha	Baseflow alpha (recession) factor	0.00 - 1.00
flo_min	Minimum aquifer storage for return flow (mm)	model default range
bf_max	Maximum baseflow rate	model default range

Calibration procedure. Parameter values were adjusted iteratively, one or a few at a time, beginning with the most sensitive parameter (the percolation coefficient), and the objective metrics were recomputed after each change until no further improvement in the monthly NSE, R-squared and PBIAS could be obtained within the permitted ranges. The final calibrated values and the resulting performance are reported in Chapter Four (Sections 4.4.2 and 4.4.3).

Sensitivity screening. Parameter sensitivity was assessed informally during this process, by observing the change in the objective metrics produced by each adjustment, rather than through a formal global sensitivity analysis. This is consistent with the manual calibration strategy adopted and with the limited number of influential parameters identified.

3.5.6 Model Evaluation Metrics and Acceptance Criteria

Model performance will be evaluated using the three metrics recommended by Moriasi *et al*. (2007) for hydrological model evaluation:

(a) Nash–Sutcliffe Efficiency (NSE):

$$NSE = 1 - [\Sigma(Q_{obs,i} - Q_{sim,i})^2 / \Sigma(Q_{obs,i} - Q_{obs,mean})^2] \quad \text{(Equation 3.5)}$$

where:

$Q_{obs,i}$ = observed discharge at time step i

$Q_{sim,i}$ = simulated discharge at time step i

$Q_{obs,mean}$ = mean of observed discharge over the evaluation period

NSE ranges from –∞ to 1.0, where NSE = 1.0 indicates perfect agreement, NSE = 0 indicates the model performs no better than the observed mean, and NSE < 0 indicates the model is worse than simply using the mean. Following Moriasi *et al*. (2007), I will consider **NSE > 0.50 as satisfactory**, **NSE > 0.65 as good**, and **NSE > 0.75 as very good** for monthly discharge simulation.

(b) Coefficient of Determination (R^2):

$$R^2 = \{[\Sigma(Q_{obs,i} - Q_{obs,mean})(Q_{sim,i} - Q_{sim,mean})]^2\} / \{\Sigma(Q_{obs,i} - Q_{obs,mean})^2 \times \Sigma(Q_{sim,i} - Q_{sim,mean})^2\} \quad \text{(Equation 3.6)}$$

R^2 ranges from 0 to 1, where values closer to 1 indicate stronger linear correlation between observed and simulated values. I will consider **R^2 > 0.60 as acceptable** for this study.

(c) Percent Bias (PBIAS):

$$PBIAS = [\Sigma(Q_{obs,i} - Q_{sim,i}) \times 100] / \Sigma Q_{obs,i} \quad \text{(Equation 3.7)}$$

PBIAS measures the average tendency of simulated values to be larger or smaller than observed values. Positive PBIAS indicates model underestimation, and negative PBIAS indicates overestimation. Following Moriasi *et al*. (2007), I will consider **|PBIAS| < 25% as satisfactory** and **|PBIAS| < 10% as very good** for monthly discharge.

3.5.7 Uncertainty Analysis Approach

Uncertainty in hydrological modelling arises from input data, model structure, and parameter estimation. Because calibration was performed by direct parameter adjustment rather than through the SUFI-2 algorithm (Section 3.5.5), no 95% prediction-uncertainty (95PPU) ensemble was generated, and the associated P-factor and R-factor statistics were not computed. The principal sources of uncertainty in this study are therefore addressed qualitatively and through independent cross-checks rather than through a formal uncertainty band: the reliance on a modelled discharge reference (GEOGLOWS) is offset by an independent evapotranspiration validation against two unrelated products (Section 3.8.1); the terrain representation is cross-checked against the HydroRIVERS network; and the sensitivity of the susceptibility classification to the AHP weights is discussed in Chapter Five. The absence of a formal uncertainty ensemble is acknowledged as a limitation, and the generation of P-factor and R-factor statistics through a subsequent SUFI-2 run is identified as further work.

3.6 Flood Susceptibility and Risk Mapping Methodology

Flood susceptibility mapping identifies areas with varying predisposition to flooding based on the integration of physical, hydrological, and anthropogenic factors (Tehrany *et al*., 2015). the study employed a GIS-based Multi-Criteria Decision Analysis (MCDA) approach using the Analytical Hierarchy Process (AHP) for factor weighting, following the framework established by Saaty (1980) and widely applied in flood susceptibility studies in Nigeria and West Africa (Adeaga *et al*., 2020; Ouma and Tateishi, 2014).

3.6.1 Susceptibility Factor Selection and Justification

The study employed eight flood susceptibility factors, each justified by its physical relationship to flood processes in Lagos State. The factor set is that used in the delivered susceptibility index and is listed in Table 3.3.

Table 3.3: Flood susceptibility factors, sources, and justifications

No.	Factor	Source	Justification
1	Elevation	Copernicus GLO-30 DEM	Lower-lying areas sit closer to the water table and lagoon level, increasing flood potential
2	Topographic Wetness Index (TWI)	Derived from DEM	Combines upslope contributing area and local slope to identify terrain prone to saturation and ponding
3	Distance to rivers / waterways	HydroRIVERS + lagoon	Proximity to water bodies increases fluvial and tidal flood exposure
4	Flow accumulation	Derived from DEM	Areas of convergent flow receive concentrated runoff from upstream
5	Runoff (SCS Curve Number)	SCS-CN from land cover and soil	Higher curve-number surfaces generate more surface runoff for a given rainfall

No.	Factor	Source	Justification
6	Slope	Derived from DEM	Flatter areas drain slowly and pond more readily
7	Extreme rainfall	CHIRPS (maximum-monthly and design rainfall), checked against NiMet	Areas receiving heavier peak rainfall have higher flood-generating potential
8	Land use / land cover	ESA WorldCover v200	Built-up, impervious surfaces raise runoff and reduce infiltration

The runoff factor (item 5) is the Soil Conservation Service Curve Number (SCS-CN) runoff estimate derived from land cover and soil as set out in Section 3.4.8 (Equations 3.2 and 3.3), not the SWAT+ streamflow output. SWAT+ covers a single representative catchment, whereas the susceptibility map is statewide, so substituting a single-catchment SWAT+ output into a statewide index would be unsound. The SWAT+ results are therefore presented in a dedicated portal panel (Section 3.7) rather than integrated into the susceptibility index, a point returned to in Chapter Five.

3.6.2 Factor Standardization Procedures

Since the eight susceptibility factors are measured in different units (metres, degrees, counts, classes, etc.), they must be standardized to a common scale before integration. the following approach was applied **linear min–max normalization** to rescale all factor values to the range [0, 1], where 0 represents the lowest susceptibility and 1 represents the highest susceptibility. The standardization formula is:

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min}) \quad \text{(Equation 3.8)}$$

where:

X_norm = standardized factor value (dimensionless, range 0 to 1)

X = raw factor value

X_min = minimum value of the factor across the study area

X_max = maximum value of the factor across the study area

For factors where higher values indicate *lower* susceptibility (e.g., elevation, slope, distance to rivers), the normalization is inverted:

X_norm = (X_max – X) / (X_max – X_min) (Equation 3.9)

This ensures that all factors are consistently oriented so that higher normalized values indicate higher flood susceptibility. For categorical factors (land cover class), ordinal susceptibility scores were assigned (0 to 1) based on expert judgment and literature (Adeaga *et al.*, 2020). For example, built-up land cover will receive a score of 1.0 (highest susceptibility due to imperviousness), forest/vegetation will receive 0.2 (moderate infiltration), and open water will be masked out.

One factor required a transformation before Equation 3.8 could be applied. Flow accumulation across the study area ranges from 1 to approximately 1.57×10^6 contributing cells and is heavily right-skewed, so a linear rescale of the raw values would map all but a small number of channel cells to approximately zero, and the factor would contribute almost nothing to the index despite carrying a weight of 0.12. The layer was therefore log-transformed as ln(1 + a), where a is the accumulated cell count, which compresses the range to approximately 0 to 14.3 before normalisation is applied. No other factor required a transformation.

3.6.3 Analytical Hierarchy Process (AHP) Weighting Methodology

The Analytical Hierarchy Process (AHP), introduced by Saaty (1980), is a structured decision technique that derives ratio-scale weights from pairwise comparisons of criteria. AHP is the most widely used MCDA weighting method in GIS-based flood susceptibility studies because of its

mathematical rigour, transparency, and built-in consistency check (Malczewski, 1999; Ouma and Tateishi, 2014; Adeaga *et al*., 2020).

Step 1 – Construct the pairwise comparison matrix. An 8 x 8 pairwise comparison matrix was constructed over the eight conditioning factors used in the delivered susceptibility index: elevation, Topographic Wetness Index (TWI), distance to rivers, flow accumulation, runoff (SCS Curve Number), slope, extreme rainfall, and land use / land cover. Each element a_ij expresses the relative importance of factor i against factor j on Saaty's 1-9 fundamental scale, with reciprocals for the reverse comparison. The judgements were informed by the literature on flood susceptibility in Lagos and comparable tropical coastal cities and by the physical understanding of flood processes in low-lying coastal terrain. The matrix is given in Table 3.4.

Table 3.4: Pairwise comparison matrix for the eight susceptibility factors

Factor	Elev	TWI	DistRiv	FlowAcc	Runoff	Slope	Rain	LULC
Elevation	1	1	1	2	2	2	3	4
TWI	1	1	1	2	2	2	2	4
Distance to rivers	1	1	1	1	1	2	2	3
Flow accumulation	1/2	1/2	1	1	1	1	2	2
Runoff (SCS-CN)	1/2	1/2	1	1	1	1	2	2
Slope	1/2	1/2	1/2	1	1	1	1	2

Factor	Elev	TWI	DistRiv	FlowAcc	Runoff	Slope	Rain	LULC
Extreme rainfall	1/3	1/2	1/2	1/2	1/2	1	1	2
Land use / land cover	1/4	1/4	1/3	1/2	1/2	1/2	1/2	1

Step 2 – Derive priority weights. Weights were obtained by the geometric-mean approximation to the principal eigenvector: for each row the geometric mean of the elements was computed and then normalised so that the weights sum to unity:

$$w_i = (a_{i1} \times a_{i2} \times \dots \times a_{in})^{(1/n)} \div \Sigma_k (a_{k1} \times a_{k2} \times \dots \times a_{kn})^{(1/n)} \quad \textit{(Equation 3.10)}$$

where:

w_i = derived weight of factor i

a_{ij} = element of the pairwise comparison matrix, expressing the importance of factor i relative to factor j

n = number of factors (8 in this study)

The derived weights, and the rounded values applied in the model, are as follows:

Factor	Derived weight	Applied weight
Elevation	0.202	0.20
Topographic Wetness Index	0.192	0.18

Factor	Derived weight	Applied weight
Distance to rivers	0.156	0.15
Flow accumulation	0.114	0.12
Runoff (SCS-CN)	0.114	0.12
Slope	0.096	0.10
Extreme rainfall	0.077	0.08
Land use / land cover	0.050	0.05

Step 3 – Consistency check. The Consistency Index (CI) and Consistency Ratio (CR) were computed as:

CI = (λmax – n) / (n – 1) (Equation 3.11)

CR = CI / RI (Equation 3.12)

where:

λmax = principal eigenvalue of the pairwise comparison matrix

n = number of factors (8 in this study)

RI = Random Index, equal to 1.41 for n = 8 (Saaty, 1980)

For the matrix in Table 3.4, λmax = 8.1139, CI = 0.0163, and **CR = 0.0115**, which is well below the acceptance threshold of 0.10. The pairwise judgements are therefore sufficiently coherent and the derived weight vector is accepted. The maximum difference between the derived and applied weights is 0.012, which is immaterial to the five-class equal-interval classification of the susceptibility index. The weighting is dominated by terrain variables: elevation and TWI together account for 0.38 of the index, and the five terrain-related factors (elevation, TWI, distance to rivers, flow accumulation, slope) together account for about 0.76. The implications of this distribution for

the map's ability to represent flooding that is not controlled by terrain position are examined in Chapter Five.

3.6.4 Flood Susceptibility Index (FSI) Calculation

The Flood Susceptibility Index is computed as the weighted linear combination of all standardized factor maps:

$$FSI = \Sigma(w_i \times X_{i,norm}) \text{ for } i = 1 \text{ to } n \quad \text{(Equation 3.13)}$$

where:

FSI = Flood Susceptibility Index (dimensionless, range 0 to 1)

w_i = AHP-derived weight for factor i (sum of all weights = 1)

$X_{i,norm}$ = standardized value (0–1) of factor i at each pixel

n = number of factors (8 in this study)

The FSI raster will be computed using raster algebra in QGIS or Python (rasterio/numpy), applying the weighted summation pixel-by-pixel across the entire Lagos State extent at 30 m resolution.

3.6.5 Classification Scheme

The continuous FSI raster (values 0 to 1) will be classified into five flood susceptibility categories using the natural breaks (Jenks) classification method, which minimizes within-class variance and maximizes between-class variance. The approximate thresholds are:

Table 3.5: Flood Susceptibility Classification Scheme

FSI Range (approx.)	**Susceptibility Class**	**Interpretation**
0.00 – 0.20	Very Low	Elevated, well-drained areas far from waterways

FSI Range (approx.)	Susceptibility Class	Interpretation
0.20 – 0.40	Low	Moderately elevated areas with some drainage limitations
0.40 – 0.60	Moderate	Transitional zones with mixed susceptibility characteristics
0.60 – 0.80	High	Low-lying areas near waterways with poor drainage and high runoff
0.80 – 1.00	Very High	Extremely low elevation, adjacent to waterways, impervious surfaces, poor soils

Note: Exact thresholds will be determined by the Jenks natural breaks algorithm applied to the computed FSI distribution and may differ from the approximate values above.

3.6.6 Flood Risk Mapping (Hazard–Exposure Integration)

Flood risk is defined as the combination of flood hazard (represented by the susceptibility map) and exposure of people, infrastructure, and assets (Ajibade *et al*., 2013). To create flood risk maps, the flood susceptibility layer was integrated with exposure indicators:

Exposure indicators: (a) Population density (from WorldPop), classified into low, moderate, high, and very high density. (b) Building density (from OpenStreetMap footprints), computed as building footprint area per hectare. (c) Road network density. (d) Presence of critical facilities (hospitals, schools, emergency services).

Risk integration: A simplified risk matrix will classify each pixel based on the combination of its susceptibility class and exposure level. For example, a pixel classified as “high susceptibility” with “high population density” will be assigned “very high risk,” while a “high susceptibility” pixel with “low exposure” will be assigned “moderate risk.” The resulting risk map will have five classes: very

low, low, moderate, high, and very high risk. This approach follows the hazard–exposure–risk framework commonly used in disaster risk assessment (UNDRR, 2019; Adeaga *et al*., 2020).

3.7 Web Portal Development Methodology

The web portal ties the project together by making the flood maps and model outputs accessible through a standard web browser to a broad range of users, from state-ministry planners to community leaders seeking to understand their area's flood risk. It allows users to explore map layers, view time series, and test rainfall scenarios through a responsive browser interface.

The architecture originally proposed for the portal (a three-tier stack comprising a PostgreSQL/PostGIS spatial database, a GeoServer tile server publishing OGC services, a FastAPI REST layer, a React single-page application, and a Celery/Redis task queue, all orchestrated under Docker) was not the architecture built. The delivered outputs are static raster and tabular products that require neither a spatial database nor a tile server, and a single-process application deploys on the available hardware without container orchestration. A lightweight architecture was therefore adopted, and the heavier stack was set aside as offering no analytical benefit for this data. This section describes the portal as built; the corresponding deviation is documented from the results side in Chapter Four (Section 4.9).

3.7.1 System Architecture

The portal is a single-process web application built with **Streamlit**, served over a single Python interface module (`model_api.py`) that loads the pre-computed raster and tabular outputs and exposes them to the interface. The interactive map is rendered with **Folium/Leaflet**. Pre-computed layers (the classified susceptibility raster, exposure summaries, SWAT+ hydrographs, scenario results, and validation outputs) are read directly from the file system as Cloud-Optimised GeoTIFFs and tabular files; no spatial database, tile server, message broker, or container orchestration layer is

used. Keeping the analytical products pre-computed means the interface performs only lightweight reads and simple on-the-fly summaries, which is well matched to the static nature of the outputs.

3.7.2 Core Portal Features

The portal delivers the five core features required by Objective (iv), organised across eight views: a project-overview page, a hazard-map view, a point-query view, an exposure dashboard, a rainfall-scenario runner, a climate panel, a SWAT+ hydrology panel, and a validation panel.

Feature 1 – Layer control with legends. The hazard-map view displays the classified five-zone susceptibility layer over a base map, with a legend, and allows the reference hydrography and exposure layers to be toggled.

Feature 2 – Time-series dashboard. The climate panel and the SWAT+ hydrology panel present rainfall and simulated discharge time series, including the observed-versus-simulated monthly hydrograph.

Feature 3 – Scenario runner. The rainfall-scenario view lets users select the predefined rainfall scenarios (baseline and the +10%, +20%, +30% multipliers) and view the resulting discharge response.

Feature 4 – Spatial query. The point-query view returns, for any clicked location, the underlying factor values and the susceptibility class at that point.

Feature 5 – Downloadable outputs. Exposure tables and scenario results can be exported by Local Government Area or sub-basin for reporting.

3.7.3 Deployment and Reproducibility

The application runs as a single Python process and is launched from the command line, with its dependencies pinned in a requirements file. The complete source code, the pre-computed output layers, and the processing scripts are held in a version-controlled repository, so the analysis and the

portal can be reproduced end to end without a bespoke server environment. This lightweight approach was chosen deliberately: because the analytical outputs are static, a reproducible single-process application meets the decision-support requirement without the operational overhead of the originally specified enterprise stack.

Software summary: Python 3.11, QGIS 3.34, GDAL 3.8, TauDEM (D8 delineation), SWAT+ (revision 62), Streamlit, Folium/Leaflet, together with the geospatial Python stack (rasterio, geopandas, numpy, xarray).

3.8 Validation and Verification Strategy

Validation demonstrates that the flood susceptibility maps and hydrological model outputs are credible and suitable for decision-making. the following approach was applied a multi-component validation strategy addressing both the hydrological model and the spatial susceptibility/risk maps.

3.8.1 Hydrologic Model Validation

SWAT+ model validation followed the split-sample approach described in Section 3.5.4. The calibrated model was run over the independent validation period (2018-2023) using historical climate inputs, and the simulated discharge was compared against the monthly GEOGLOWS reference using NSE, R-squared and PBIAS (Equations 3.5-3.7). The acceptability criteria of Section 3.5.6 (NSE > 0.50, R-squared > 0.60, |PBIAS| < 25%) were required over the validation period for the model to be considered reliable for scenario analysis (Moriasi *et al.*, 2007).

Two further validation checks were applied. First, observed and simulated monthly hydrographs were compared visually to assess reproduction of the seasonal pattern (reported in Chapter Four). Second, and in place of a comparison of annual water-balance components against literature values, simulated evapotranspiration was validated independently against two unrelated gridded evapotranspiration products, the MODIS MOD16A2GF retrieval and the TerraClimate water-balance product. Because evapotranspiration formed no part of the calibration objective, this

constitutes an out-of-sample test of the model's water balance that is stronger than a comparison against published figures, and it introduces a data source wholly unconnected to the GEOGLOWS discharge reference used for calibration. No comparison of simulated peak-discharge timing against LASEMA event dates was undertaken; such an event-timing comparison is identified as further work.

3.8.2 Spatial Susceptibility and Risk Map Validation

The flood susceptibility map was validated against the flood inventory datasets of Section 3.3.7. The primary validation surface was the continuous MODIS 2012 flood *extent* rather than a set of discrete flood points; validating against a continuous observed extent is a more demanding test than validating against a small number of points, because it requires discrimination across every cell of the observed footprint.

Step 1 – Overlay analysis. The observed flood extent was overlaid on the classified susceptibility map, and the share of observed flooding falling in each susceptibility class was computed and expressed as an enrichment ratio (the share of observed flooding in a zone divided by that zone's share of the state area). A valid susceptibility map should concentrate observed flooding in the higher classes, giving enrichment ratios above one in the High and Very High zones.

Step 2 – Area under the ROC curve (AUC-ROC). The AUC-ROC measures the discriminatory power of the susceptibility model (Tehrany *et al*., 2015). Continuous FSI values were extracted at flooded cells (positive cases) and at an equal number of randomly sampled non-flooded cells (negative cases); the ROC curve was constructed by plotting the true-positive rate against the false-positive rate across FSI thresholds, and the area under the curve was computed. AUC values are interpreted as 0.50 = no discrimination, 0.70-0.80 = acceptable, 0.80-0.90 = excellent, and > 0.90 = outstanding. Consistent with Table 3.6, an AUC-ROC above 0.70 was taken as the threshold for acceptable discrimination.

Step 3 – Confusion-matrix analysis. At the FSI threshold maximising the Youden index (J = sensitivity + specificity – 1), each cell was classified as predicted flood or predicted non-flood, and a confusion matrix was constructed. Sensitivity, specificity, overall accuracy and precision were then computed as given in Equations 3.14 to 3.17:

Sensitivity = TP / (TP + FN) (Equation 3.14)

Specificity = TN / (TN + FP) (Equation 3.15)

Overall accuracy = (TP + TN) / (TP + TN + FP + FN) (Equation 3.16)

Precision = TP / (TP + FP) (Equation 3.17)

where:

TP = true positives, flooded cells correctly predicted as flood-prone

TN = true negatives, non-flooded cells correctly predicted as not flood-prone

FP = false positives, non-flooded cells predicted as flood-prone

FN = false negatives, flooded cells predicted as not flood-prone

A sensitivity of at least 0.70 was set as the target, meaning that the map should correctly identify at least 70% of flooded cells as High or Very High susceptibility.

Local Government Area cross-check. As a secondary, non-spatial check, NEMA 2022 report counts were compared at Local Government Area level against modelled exposure using Spearman rank correlation. Because the NEMA records were available only at LGA level (Section 3.3.7), this check is coarser than the pixel-level extent validation, and it is reported in Chapter Four with the confounding by population made explicit.

3.8.3 Accuracy Assessment Metrics Summary

Table 3.6 summarises the validation metrics, their acceptance criteria, and the data sources used for evaluation.

Table 3.6: Summary of validation metrics and acceptance criteria

Component	Metric	Acceptable value	Data source
SWAT+ model	NSE	> 0.50 (satisfactory)	GEOGLOWS monthly discharge
SWAT+ model	R-squared	> 0.60	GEOGLOWS monthly discharge
SWAT+ model	PBIAS	|PBIAS| < 25%	GEOGLOWS monthly discharge
SWAT+ model	P-factor	>= 0.70	Not assessed - no uncertainty ensemble generated (Section 3.5.7)
SWAT+ model	R-factor	<= 1.5	Not assessed - no uncertainty ensemble generated (Section 3.5.7)
SWAT+ model	Evapotranspiration PBIAS	|PBIAS| < 25%	MODIS MOD16A2GF and TerraClimate (independent)
Susceptibility map	AUC-ROC	> 0.70 (acceptable)	MODIS 2012 flood extent
Susceptibility map	Sensitivity	>= 0.70	MODIS 2012 flood extent
Susceptibility map	Overall accuracy	> 0.70	Flood and non-flood

Component	Metric	Acceptable value	Data source
			cells
Susceptibility map	Flood-in-high-class %	> 60%	Overlay analysis

CHAPTER FOUR

RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the results obtained from the hydrological modelling, flood susceptibility mapping, and validation procedures described in Chapter Three. The results are analysed and compared against the acceptance criteria defined in Table 3.6 to evaluate whether each component of the study meets the performance standards set for it. The chapter is organized to present the assembled data stack, the SWAT+ model setup and calibration results, scenario simulation results, independent evapotranspiration validation, flood susceptibility and exposure results, spatial validation of the susceptibility map, the developed web portal, and a summary of attainment against all acceptance criteria.

4.2 Data Acquisition and Preprocessing Results

Below are the results of the data acquisition and harmonisation procedures carried out before modelling;

4.2.1 Final Assembled Data Stack

Table 4.1 Final data stack and deviations from proposed sources

Layer	Source used	Resolution	Deviation from Section 3.3
Rainfall (SWAT forcing)	CHIRPS daily	5 km, 23 stations	None
Temperature	ERA5-Land tmax/tmin	9 km	None

Layer	Source used	Resolution	Deviation from Section 3.3
Rainfall (statistics)	NiMet Ikeja, 1996–2025	Station	None
DEM	Copernicus GLO-30	30 m → 90 m	SRTM replaced
Rivers and basins	HydroRIVERS v1.0	Vector	None
Land cover	ESA WorldCover v200	10 m → 90 m	None
Soil	SoilGrids 0–5 cm sand/clay	250 m → 90 m	Single layer only
Discharge	GEOGLOWS v2, reach 140585128	Daily, 1940–2026	NIHSA replaced
Flood inventory	MODIS 2012 extent; NEMA 2022	250 m / LGA	Compiled from open archives
Population	WorldPop	100 m	None

All ten required layers were successfully acquired, reprojected to EPSG:32631, and clipped to their respective study extents with no gaps in coverage. The success criterion stated in Objective (i) is therefore satisfied.

Two of the ten layers were obtained from sources other than those originally proposed. The Shuttle Radar Topography Mission DEM was replaced with the Copernicus GLO-30 DEM because the available SRTM tiles terminated at 7°N and did not cover the northern headwaters of the Ogun basin, which extend to approximately 8.95°N. The Copernicus product offers equivalent 30 m resolution with complete basin coverage.

Observed discharge from the Nigeria Hydrological Services Agency was not obtained within the project period. GEOGLOWS ECMWF retrospective discharge at reach 140585128, located at Kara/Akute on the Ogun main stem, was adopted in its place. This series comprises 31,589 daily values covering 1940 to 2026, aggregated to monthly means for model evaluation.

4.3 SWAT+ Model Setup Results

4.3.1 Catchment Delineation

Table 4.2 Catchment delineation results

Property	Value
Delineated catchment area	19,915 km²
Number of subbasins	23
Number of channels	71
Number of landscape units	71
Number of HRUs	243 (from 1,971 candidates)
Outlet location	Kara/Akute, UTM 31N 542139 E, 736226 N
HydroRIVERS reference area	24,816 km²

Delineation produced a catchment area of 19,915 km², which compares with an independent HydroRIVERS upstream-area estimate of 24,816 km² for the Ogun basin as a whole. The delineated extent therefore represents the portion of the Ogun system draining to the Kara/Akute outlet above the Lagos Lagoon.

An initial delineation performed without stream burn-in produced a maximum flow accumulation of only 462 km², approximately 2% of the expected catchment area. Inspection confirmed the DEM contained no voids and had a valid elevation range of 0 to 585 m, indicating that the failure arose

from flow dispersion across the flat valley terrain rather than from data gaps. Burning in the HydroRIVERS network and relocating the outlet to the nearest main-stem vertex raised maximum flow accumulation to 20,592 km², producing the delineation reported above.

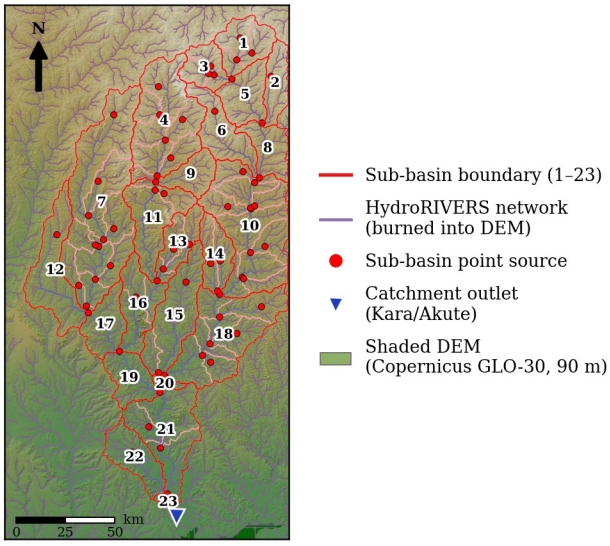


Figure 3: Delineated Ogun catchment showing sub-basins, the burned-in HydroRIVERS stream network and the Kara/Akute outlet

4.3.2 HRU Input Characteristics

Table 4.3 Basin land cover and soil composition

Land cover class	% of basin	Soil texture	% of basin
Tree cover	43.0	Sandy loam	88.0
Shrubland	23.0	Sandy clay loam	11.0
Cropland	16.0	Other	1.0
Grassland	12.0		
Built-up	4.5		

The basin is predominantly vegetated, with tree cover and shrubland together accounting for 66% of the area and built-up surfaces occupying only 4.5%. Soils are dominated by two textures, with sandy loam covering 88% of the basin. The low variability in soil texture permitted representation through a custom user-soil database containing twelve texture classes built on the FAO schema.

4.4 SWAT+ Calibration and Validation Results

4.4.1 Baseline Uncalibrated Performance

Table 4.4 Uncalibrated model performance against GEOGLOWS monthly discharge, 2011–2023

Metric	Value	Criterion (Table 3.6)	Status
NSE	−0.10	> 0.50	Not met
R^2	0.51	> 0.60	Not met
PBIAS	−48.9%	< ±25%	Not met
Mean simulated	337 m^3/s	—	—
Mean observed	226 m^3/s	—	—

The uncalibrated model failed all three acceptance criteria. The combination of a moderate R^2 of 0.51 with a strongly negative NSE indicates that the model reproduced the seasonal shape of the hydrograph while partitioning water incorrectly. Simulated wet-season peaks were substantially overestimated, while dry-season baseflow was severely underestimated, with simulated January to March flows of approximately 4 m^3/s against observed values near 50 m^3/s. Calibration was therefore required before the model could be used for scenario analysis.

4.4.2 Calibrated Parameter Set

Table 4.5 Final calibrated parameter values

Parameter	Description	Value	Permitted range
perco	Percolation coefficient	0.30	0.00–1.00
petco	PET multiplier	1.20	0.80–1.20
canmx	Maximum canopy	12.0	0.00–100.0

Parameter	Description	Value	Permitted range
	storage (mm)		
alpha	Baseflow alpha factor	0.022	0.00–1.00
flo_min	Minimum aquifer storage for return flow (mm)	10.0	—
bf_max	Maximum baseflow rate	0.55	—

Six parameters were adjusted from their defaults. Reducing the percolation coefficient from its default of 0.90 to 0.30 produced the first substantial improvement, raising NSE from −0.10 to 0.18, and identified the catchment as baseflow-dominated with surface runoff contributing approximately 18 mm/yr. Raising the PET multiplier to 1.20 moved PBIAS from −30% to +3%. Canopy storage of 12 mm reduced excessive wet-season peaks, and the three aquifer parameters restored dry-season baseflow.

petco reached 1.20, the maximum value permitted by the SWAT+ parameter database. The implications of this constraint are examined in Section 4.6.3.

The soil evaporation compensation factor, plant uptake compensation factor and available water capacity were tested during calibration and were found to be either inert or to move the simulation in the wrong direction in this SWAT+ revision. They were therefore excluded from the final parameter set.

4.4.3 Calibrated Model Performance

Table 4.6 Calibrated SWAT+ performance against GEOGLOWS monthly discharge

Period	NSE	R^2	PBIAS	Criterion	Status
Calibration, 2011–2017	0.41	0.56	+6.1%	NSE > 0.50, R^2 > 0.60, PBIAS < ±25%	Partially met
Validation, 2018–2023	0.57	0.64	−3.2%	NSE > 0.50, R^2 > 0.60, PBIAS < ±25%	Met
Full period, 2011–2023	0.507	0.606	+1.6%	NSE > 0.50, R^2 > 0.60, PBIAS < ±25%	Met

The calibrated model satisfies all three acceptance criteria over the full evaluation period, with NSE of 0.507, R^2 of 0.606 and PBIAS of +1.6%. Objective (ii) is therefore met. Mean simulated discharge of 222 m^3/s compares with 226 m^3/s observed.

Validation-period performance exceeds calibration-period performance on all three metrics. This indicates that the parameter set is not over-fitted to the calibration window.

The full-period margins above threshold are narrow. NSE exceeds its criterion by 0.007 and R^2 by 0.006. The model is therefore satisfactory but only marginally so, and any subsequent change to forcing data or parameter values would require the evaluation to be repeated.

4.4.4 Monthly Hydrograph Comparison

Table 4.7 Observed and simulated monthly discharge climatology (m^3/s), 2011–2023

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Observed	69	47	42	59	90	184	328	332	684	518	235	116
Simulated	44	26	14	13	23	115	312	363	760	720	196	72

The model reproduces the timing of the flood season, including the September maximum that drives fluvial discharge into the Lagos Lagoon. Two discrepancies are evident. October discharge is overestimated at 720 m^3/s against 518 m^3/s observed, indicating that recession from the wet season is too slow. The February to May transition remains underestimated, indicating that the early rains generate insufficient flow in the simulation.

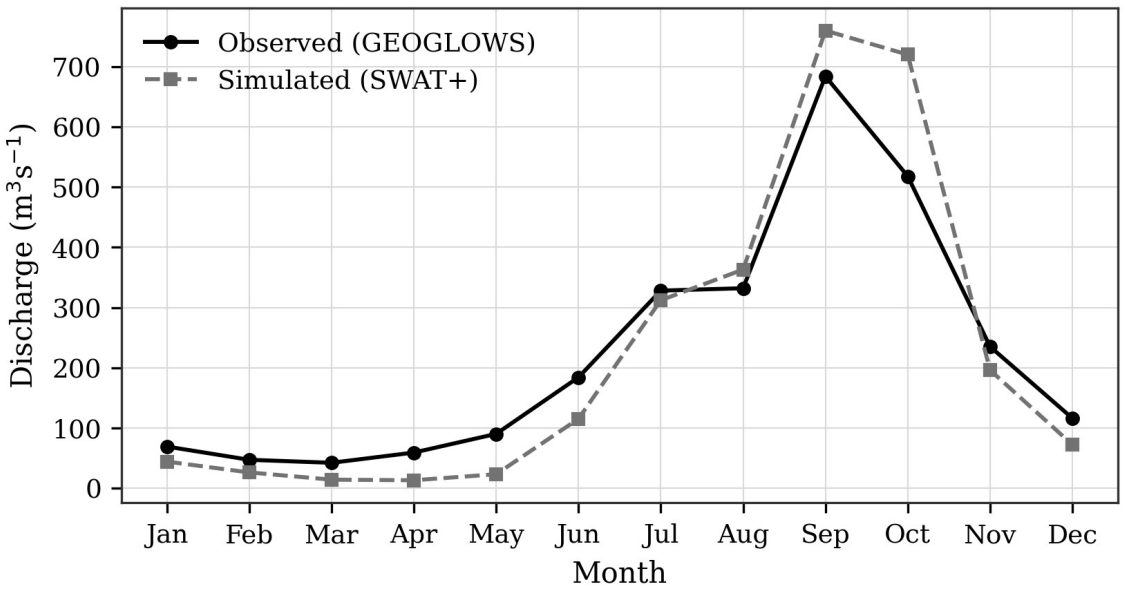


Figure 4: Observed and simulated monthly discharge climatology at the Kara/Akute outlet, 2011–2023 (data from Table 4.7)

4.5 Scenario Simulation Results

Table 4.8 Simulated outlet discharge under rainfall scenarios

Scenario	Mean (m³/s)	Change in mean	Peak month (m³/s)	Change in peak
Baseline	222	—	1420	—
+10% rainfall	257	+15.5%	1558	+9.7%
+20% rainfall	292	+31.3%	1686	+18.7%
+30% rainfall	328	+47.6%	1821	+28.2%

Catchment discharge responds non-linearly to increased rainfall. A 10% increase in rainfall produces a 15.5% increase in mean discharge, and a 30% increase produces a 47.6% increase. The amplification factor is therefore approximately 1.55 across the range tested.

Peak-month discharge responds less strongly than mean discharge, rising by 28.2% under a 30% rainfall increase. This indicates that the additional rainfall volume is distributed across the wet season rather than concentrated at the seasonal maximum.

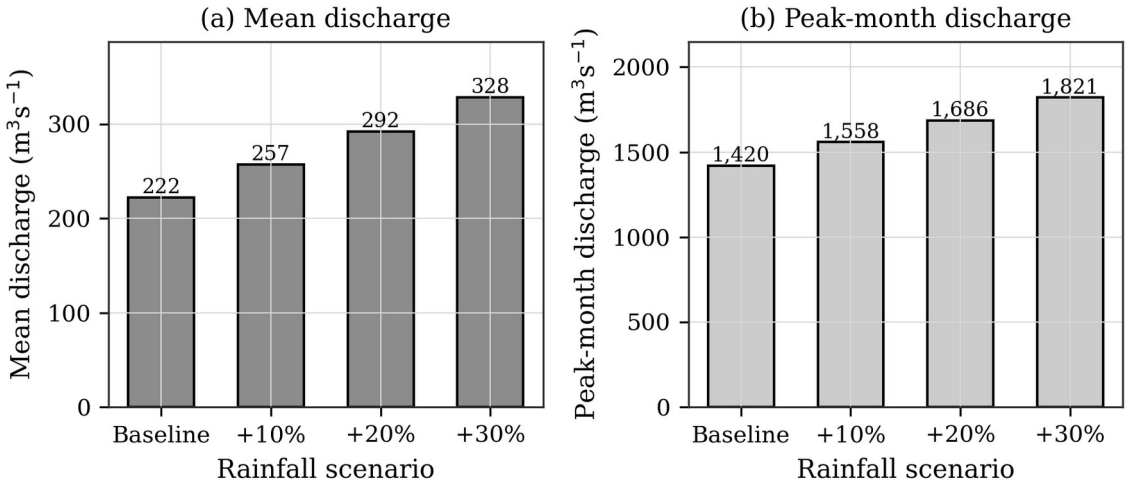


Figure 5: Simulated (a) mean and (b) peak-month outlet discharge under the rainfall scenarios (data from Table 4.8)

Table 4.9 Extreme historical scenario results

Scenario	Annual mean (m³/s)	Peak month (m³/s)
Wettest simulated year (2019)	308	1053
2022 flood year	Near baseline	Near baseline

The wettest year in the simulation period was 2019, producing an annual mean of 308 m³/s. The 2022 flood year, which produced severe flooding nationally, returned near-baseline discharge in the rainfall-only simulation. This is consistent with documented evidence that the 2022 event in this basin was driven substantially by releases from Oyan Dam rather than by rainfall alone. As reservoir operation is not represented in the model, the 2022 event falls outside its representational scope.

The three scenario families required by Objective (ii), comprising baseline, extreme historical and climate sensitivity, were therefore produced.

4.6 Independent Evapotranspiration Validation Results

4.6.1 Basis of the Validation

The calibration reported in Section 4.4 used GEOGLOWS discharge for both the calibration and validation periods, separated only in time. To provide a test independent of that data source, simulated evapotranspiration was compared against two unrelated reference products. Evapotranspiration entered no part of the calibration objective function, so agreement constitutes an out-of-sample test rather than a fitted result.

4.6.2 Comparison Against Reference Products

Table 4.10 Simulated evapotranspiration against independent references, 155 months

Reference	NSE	R^2	PBIAS	Criterion	Status
MODIS MOD16A2 GF	0.14	0.36	+14.4%	PBIAS < ±25%	Met
TerraClimate AET	0.18	0.39	+12.5%	PBIAS < ±25%	Met

Table 4.11 Annual evapotranspiration totals

Source	Annual mean (mm/yr)
MODIS MOD16A2GF	920
TerraClimate AET	900
SWAT+ simulated	787

Annual evapotranspiration volume falls within the ±25% acceptance band against both references, at +14.4% and +12.5% respectively. Monthly agreement is weak, with NSE of 0.14 and 0.18.

The two reference products agree with each other to within 20 mm/yr, or approximately 2%, despite being derived by entirely different methods. MODIS MOD16A2GF is a satellite retrieval, while TerraClimate is a climatic water-balance product. Their agreement indicates that the discrepancy with the simulation is a property of the model rather than an artefact of a single reference dataset.

Table 4.12 Monthly evapotranspiration climatology (mm), MODIS against simulated

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
MODIS	27.8	32.1	53.7	65.0	87.0	101.7	110.2	112.3	113.6	88.5	83.2	41.9
Simulated	55.9	49.8	59.5	64.9	75.6	68.8	67.9	65.4	70.7	73.8	72.0	63.0

MODIS evapotranspiration varies across an annual range of approximately 86 mm, while the simulated series varies across approximately 26 mm. The simulated seasonal amplitude is therefore approximately one third of that observed. The model produces excess evapotranspiration during the dry season and a deficit at the wet-season maximum, while returning approximately the correct annual total.

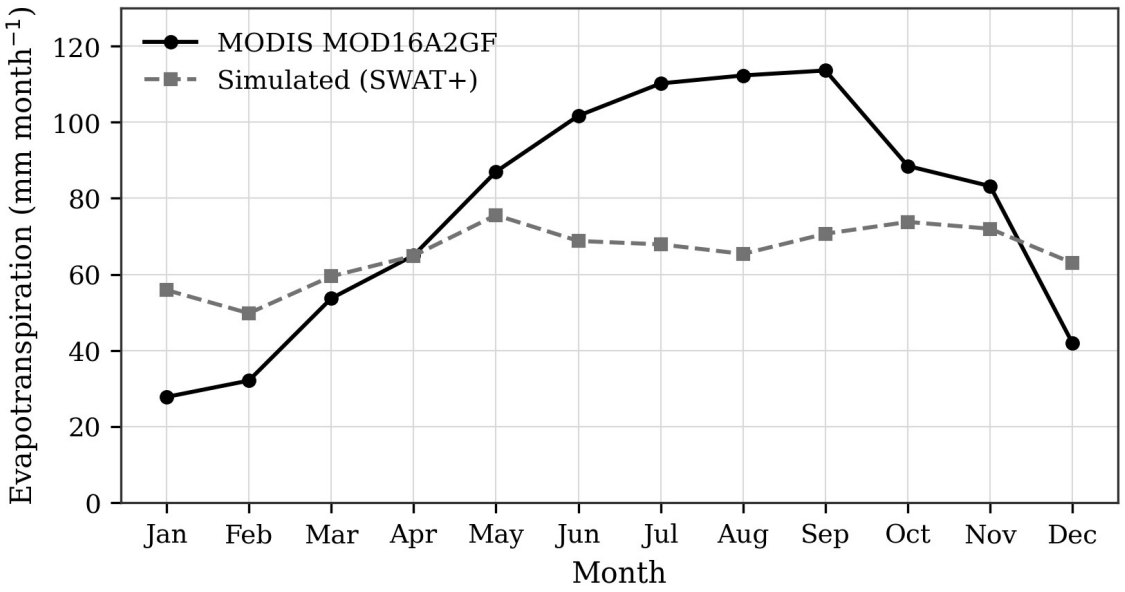


Figure 6: MODIS MOD16A2GF and simulated monthly evapotranspiration climatology (data from Table 4.12)

4.6.3 Multi-Variable Calibration Test

Table 4.13 Simultaneous discharge and evapotranspiration performance

petco	Discharge NSE	Discharge R^2	Discharge PBIAS	ET NSE	ET PBIAS	Meets Table 3.6
1.20 (adopted)	0.507	0.606	+1.6%	0.14	+14.4%	Yes
1.35	0.530	0.596	+15.7%	0.30	+6.6%	No
1.50	0.507	0.583	+28.7%	0.37	+1.1%	No

Evapotranspiration performance improves monotonically as the PET multiplier is raised, with ET NSE rising from 0.14 to 0.37, while discharge performance falls below the acceptance criteria. At petco of 1.35 the discharge R^2 of 0.596 falls below the criterion of 0.60. At petco of 1.50 both R^2 and PBIAS fail.

The two objectives are therefore in direct conflict within the available parameter space, and the adopted parameter set of petco 1.20 is the only value tested that satisfies the discharge criteria. It was retained on that basis. Values of 1.35 and 1.50 were reachable only by temporarily raising the permitted parameter ceiling, which was restored afterwards.

4.7 Flood Susceptibility Mapping Results

4.7.1 Pairwise Comparison Matrix and Consistency Check

Table 4.14 Pairwise comparison matrix for the eight susceptibility factors

Factor	Elev	TWI	DistRiv	FlowAcc	Runoff	Slope	Rain	LULC
Elevation	1	1	1	2	2	2	3	4
TWI	1	1	1	2	2	2	2	4
Distance to rivers	1	1	1	1	1	2	2	3
Flow accumulation	1/2	1/2	1	1	1	1	2	2
Runoff	1/2	1/2	1	1	1	1	2	2
Slope	1/2	1/2	1/2	1	1	1	1	2
Extreme rainfall	1/3	1/2	1/2	1/2	1/2	1	1	2
Land use / land	1/4	1/4	1/3	1/2	1/2	1/2	1/2	1

Factor	Elev	TWI	DistRiv	FlowAcc	Runoff	Slope	Rain	LULC

cover

Table 4.15 Derived AHP weights, applied weights, and consistency statistics

Factor	Derived weight	Applied weight	Difference
Elevation	0.202	0.20	+0.002
Topographic Wetness Index	0.192	0.18	+0.012
Distance to rivers	0.156	0.15	+0.006
Flow accumulation	0.114	0.12	−0.006
Runoff (SCS-CN)	0.114	0.12	−0.006
Slope	0.096	0.10	−0.004
Extreme rainfall	0.077	0.08	−0.003
Land use and land cover	0.050	0.05	0.000
Total	**1.000**	**1.00**	—

Consistency statistic	Value	Criterion	Status
λ_max	8.1139	—	—
Consistency Index (CI)	0.0163	—	—
Random Index	1.41	—	—

Consistency statistic	Value	Criterion	Status
(RI), n = 8			
Consistency Ratio (CR)	**0.0115**	**≤ 0.10**	**Met**

The pairwise matrix yields a Consistency Ratio of 0.0115, which is well below the acceptance criterion of 0.10 stated in Section 3.6.3. The pairwise judgements are therefore sufficiently coherent, and the derived weight vector is accepted. Weights were obtained by the geometric mean approximation to the principal eigenvector, as specified in Section 3.6.3, and λ_max was computed from the resulting weight vector.

The weights applied in the model were rounded to two decimal places, as recorded in the model configuration. The maximum difference between the derived and applied values is 0.012, occurring for the Topographic Wetness Index, and arises because pairwise judgements are restricted to Saaty's integer scale and its reciprocals. A difference of this magnitude is immaterial to the classification, since the susceptibility index is subsequently divided into five equal-interval classes.

The weighting is dominated by terrain variables. Elevation and the Topographic Wetness Index together account for 0.394 of the index, and the five terrain-related factors, comprising elevation, TWI, distance to rivers, flow accumulation and slope, together account for 0.760. The two rainfall-driven components, runoff and extreme rainfall, account for 0.191, and land use and land cover for 0.050.

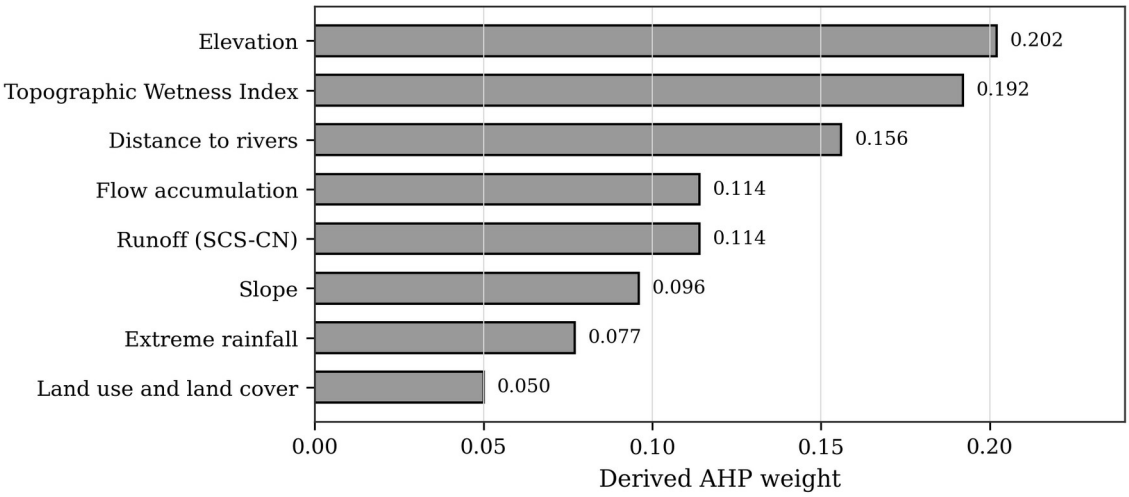


Figure 7: AHP-derived weights of the eight flood conditioning factors, consistency ratio 0.0115 (data from Table 4.15)

The implications of this distribution for the map's ability to represent flooding that is not controlled by terrain position are examined in Section 5.1.

4.7.2 Susceptibility Classification and Population Exposure

Table 4.16 Susceptibility zone areas and resident population, Lagos State

Zone	Area (km²)	% of Lagos	Population	% of population
Very Low	0.8	0.02	10,077	0.08
Low	168.7	4.46	692,475	5.58
Moderate	1,762.9	46.61	6,908,435	55.62
High	1,055.9	27.92	4,368,569	35.17
Very High	793.7	20.99	441,500	3.55
Total	3,782.0	100.00	12,421,056	100.00

An estimated 4,810,069 residents, or 38.7% of the state population, fall within the High or Very High susceptibility zones. The High and Very High zones together cover 48.91% of the state area.

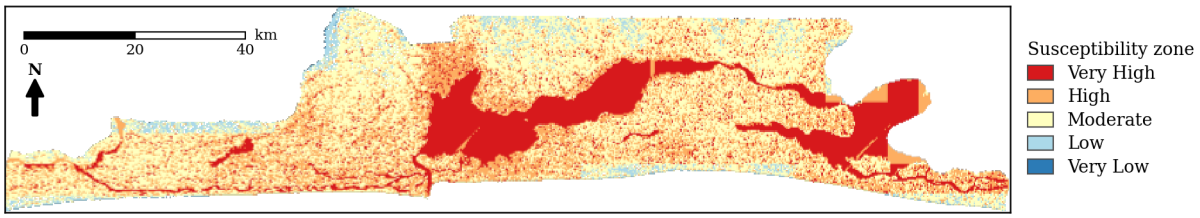


Figure 8: Flood susceptibility map of Lagos State classified into five zones

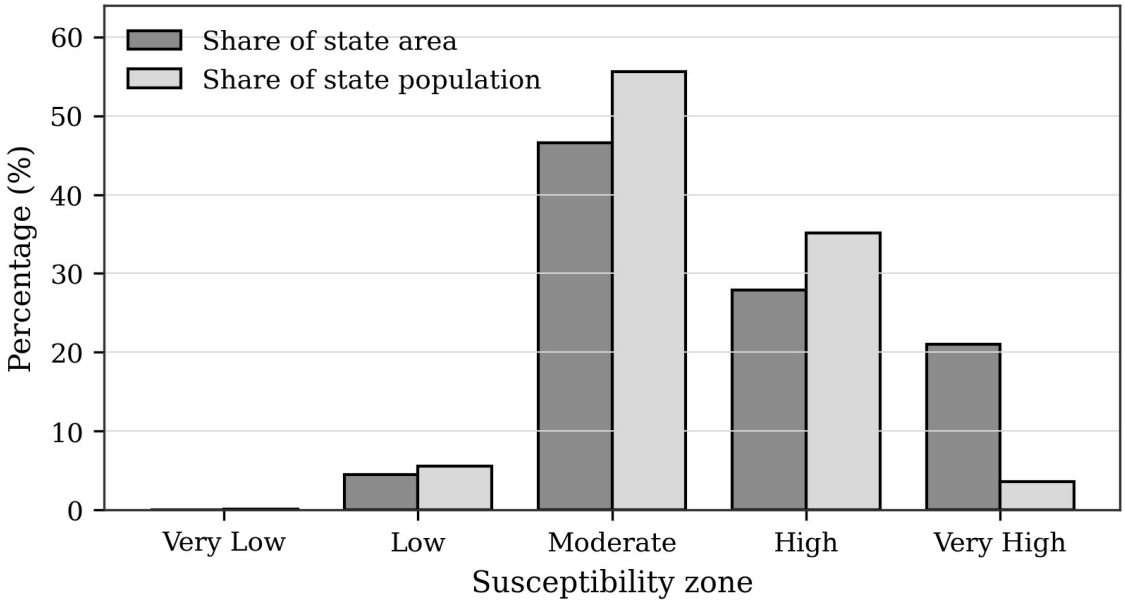


Figure 9: Share of state area and of resident population by susceptibility zone (data from Table 4.16)

Table 4.17 Highest and lowest LGA exposure

LGA	% population in High or Very High
Eti-Osa	67.8
Lagos Island	65.0
Shomolu	58.2
Ojo	53.4
Ikeja	10.9
Agege	9.9
Ifako/Ijaye	7.2

Exposure ranges from 67.8% in Eti-Osa to 7.2% in Ifako/Ijaye. The highest-exposure LGAs lie along the coastal and lagoon-adjacent low-lying belt, which is consistent with the terrain-dominated weighting reported in Table 4.14.

4.7.3 Rainfall Frequency Analysis

Table 4.18 Gumbel design rainfall, NiMet Ikeja station, 1996–2025

Return period (years)	2	5	10	25	50	100
Design rainfall (mm)	1,734	2,205	2,516	2,909	3,201	3,490

The design rainfall was derived from a Gumbel distribution fitted to the annual series, with location parameter 1582.4 and scale parameter 414.8. Mean annual rainfall over the record is 1,821.8 mm.

The observed maximum of 3,931.7 mm, recorded in 2018, corresponds to a return period of approximately 289 years on this fit. This indicates either an exceptional event or that the Gumbel distribution understates the upper tail at this station.

Mann-Kendall testing found no significant trend in annual rainfall over the record, with Z of 0.18 and p of 0.86. A significant increasing trend was found in mean annual temperature, with Z of 4.17, p below 0.001, and a slope of +0.44 °C per decade.

4.8 Spatial Validation Results

4.8.1 Validation Datasets

Two independent observational datasets were compiled from open archives, as no comprehensive spatial flood database exists for Lagos State.

Table 4.19 Flood validation datasets

Dataset	Source	Coverage in Lagos
MODIS 250 m flood extent, Jul–Nov 2012	NASA/DFO via HDX	35.2 km² across 290 patches
NEMA 2022 flood records	NEMA via HDX	62 reports across 13 of 20 LGAs

The 2012 MODIS product provides 35.2 km² of detected flood extent within Lagos State, distributed across 290 discrete patches. Permanent water bodies are masked in the source product, so the extent represents detected flooding rather than the lagoon system.

No geocoding of NEMA community names was attempted. The records carry community names but no coordinates, and assigning coordinates from place names would introduce fabricated locations into the dataset used for validation. The NEMA records were therefore analysed only at the LGA level of aggregation that the source supports.

4.8.2 Overlay Analysis

Table 4.20 Distribution of observed 2012 flood extent across susceptibility zones

Zone	% of observed flood	% of Lagos area	Enrichment ratio
Very Low	0.00	0.02	0.00
Low	2.32	4.46	0.52
Moderate	30.61	46.61	0.66
High	33.19	27.92	1.19
Very High	33.89	20.99	1.61
High + Very High	67.08	48.91	1.37

The enrichment ratio, calculated as the share of observed flooding in a zone divided by that zone's share of the state area, rises monotonically across the five susceptibility classes. Observed flooding is 1.61 times more concentrated in the Very High zone than would occur by chance, and is depleted in the Low and Moderate zones at ratios of 0.52 and 0.66 respectively.

The proportion of observed flooding falling in the High and Very High classes is 67.08%, against the acceptance criterion of greater than 60% stated in Table 3.6. This criterion is met.

The High and Very High zones together cover 48.91% of the state. A randomly assigned classification would therefore place approximately 49% of observed flooding within them, and the measured 67.08% represents an improvement of 18 percentage points over chance.

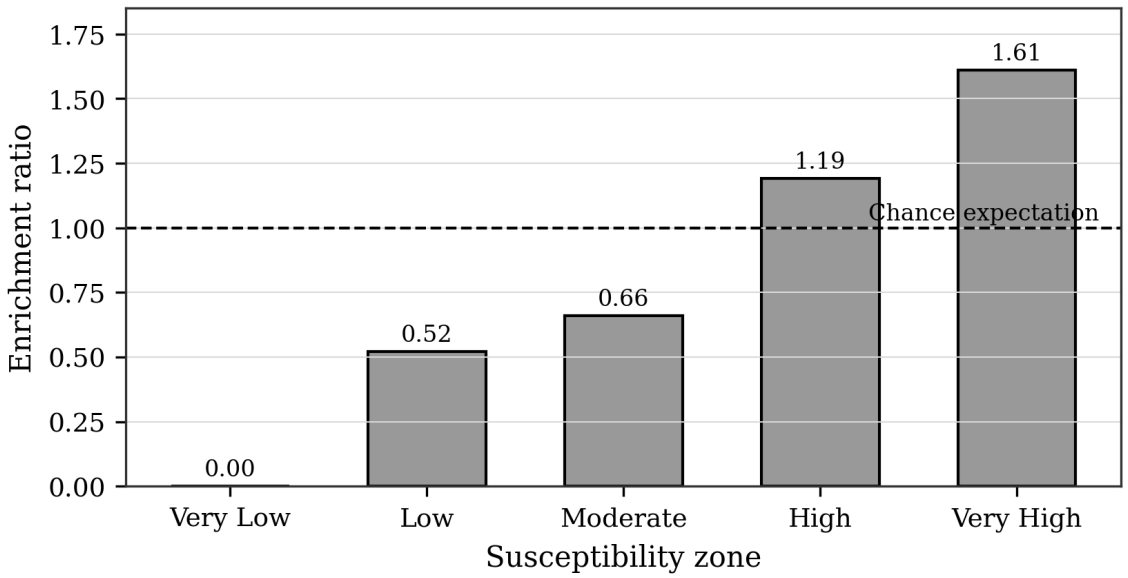


Figure 10: Enrichment ratio of the observed 2012 flood extent by susceptibility zone (data from Table 4.20)

4.8.3 Discriminatory Power and Confusion Matrix

Table 4.21 Susceptibility map discriminatory performance

Metric	Value	Criterion (Table 3.6)	Status
AUC-ROC	0.62	> 0.70	Not met
Sensitivity	0.685	≥ 0.70	Not met
Specificity	0.494	—	—
Overall accuracy	0.526	> 0.70	Not met
Precision	0.213	—	—
Flood-in-high class	67.08%	> 60%	Met

Table 4.22 Confusion matrix at the Youden-optimal threshold, FSI = 0.596

	Predicted flood	Predicted non-flood
Observed flood	26,799	12,309
Observed non-flood	98,888	96,652

The susceptibility map meets one of the four acceptance criteria set for it. The AUC-ROC of 0.62 falls below the criterion of 0.70, though it exceeds 0.50, which would indicate no discrimination. Sensitivity of 0.685 falls marginally below the criterion of 0.70, and overall accuracy of 0.526 falls substantially below it.

Specificity of 0.494 and precision of 0.213 identify the cause. Because the High and Very High zones occupy 48.91% of the state, they capture flooded and non-flooded ground at approximately the same rate. Of 125,687 cells predicted as flood-prone at the optimal threshold, 98,888 were not flooded in the 2012 event.

Objective (v) is therefore only partially attained. The SWAT criterion is satisfied, while the susceptibility criteria are not.

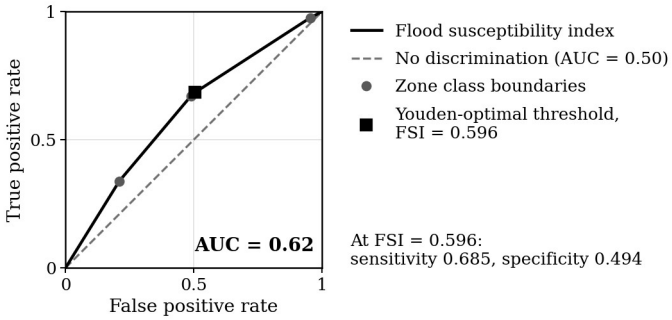


Figure 11: Receiver operating characteristic curve for the flood susceptibility index against the 2012 MODIS flood extent, AUC = 0.62

4.8.4 LGA-Level Comparison with NEMA Records

Table 4.23 Spearman rank correlation between NEMA report counts and modelled quantities

Compared quantity	Spearman ρ
Modelled % population in High or Very High	−0.45
Modelled absolute population exposed	+0.17
Raw LGA population	+0.57

NEMA report counts correlate more strongly with raw LGA population (ρ of +0.57) than with either modelled exposure quantity. The Mann-Whitney comparison between LGAs with and without reports was not statistically significant, at p of 0.10.

This indicates that the NEMA dataset predominantly records where population and reporting capacity are concentrated rather than where flooding occurs. The dataset is therefore confounded and cannot validate the susceptibility map in either direction. This test is reported as inconclusive.

Table 4.24 LGAs with highest NEMA report counts against modelled exposure

LGA	NEMA reports	Modelled % High or Very High
Alimosho	11	14.4

LGA	NEMA reports	Modelled % High or Very High
Ikorodu	10	22.4
Ojo	10	53.4
Epe	7	23.5
Kosofe	6	47.1
Agege	5	9.9

Three of the six LGAs generating the most flood reports — Alimosho, Ikorodu and Agege — rank among the lowest in modelled exposure at 14.4%, 22.4% and 9.9% respectively. These are dense inland local government areas.

4.9 Web Portal Development Results

Table 4.25 Delivered portal features against Objective (iv)

Required feature	Delivered	Implementation
Layer control with legends	Yes	Hazard map view with five-zone legend
Time-series dashboard	Yes	Climate panel and SWAT+ hydrograph
Scenario runner	Yes	Rainfall scenario view
Spatial query popups	Yes	Point query view
Downloadable outputs	Yes	Exposure tables and scenario results

All five core features required by Objective (iv) were delivered. The portal provides eight views: project overview, hazard map, point query, exposure dashboard, rainfall scenario runner, climate panel, SWAT+ hydrology panel, and validation panel.

The implementation stack differs from that specified in Objective (iv). The portal was built as a Streamlit application over a single Python interface module, with Folium and Leaflet providing the interactive map. PostgreSQL/PostGIS, GeoServer, FastAPI, React and Docker were not used. The functional requirements of Objective (iv) are therefore satisfied, while the architectural specification is not.

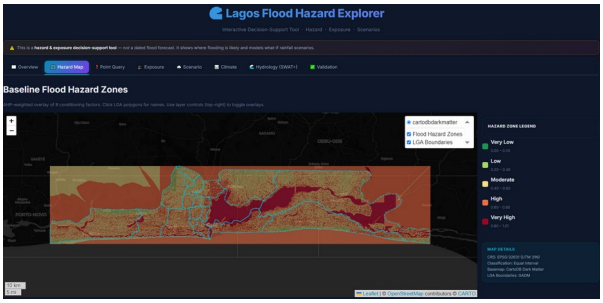


Figure 12: Portal hazard map view showing the five-zone susceptibility legend

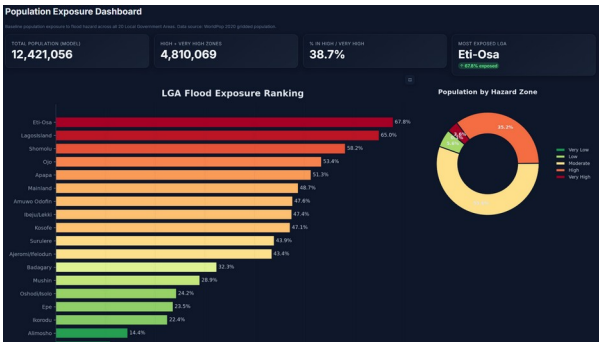


Figure 13: Portal exposure dashboard showing population by susceptibility zone and LGA

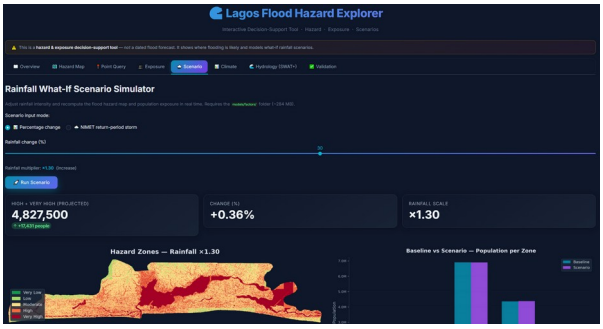


Figure 14: Portal rainfall scenario runner

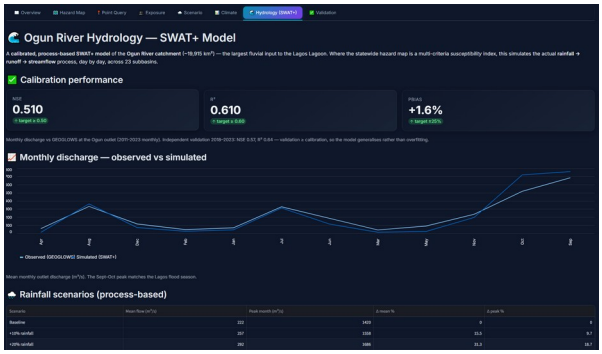


Figure 15: Portal SWAT+ hydrology panel

4.10 Summary of Attainment Against Acceptance Criteria

Table 4.26 Attainment against the acceptance criteria of Table 3.6

Component	Metric	Criterion	Achieved	Status
SWAT+	NSE	> 0.50	0.507	Met
SWAT+	R^2	> 0.60	0.606	Met

Component	Metric	Criterion	Achieved	Status
SWAT+	PBIAS	< ±25%	+1.6%	Met
SWAT+	P-factor	≥ 0.70	Not computed	Not assessed
SWAT+	R-factor	≤ 1.5	Not computed	Not assessed
Susceptibility	AUC-ROC	> 0.70	0.62	Not met
Susceptibility	Sensitivity	≥ 0.70	0.685	Not met
Susceptibility	Overall accuracy	> 0.70	0.526	Not met
Susceptibility	Flood-in-high class	> 60%	67.08%	Met

Four of the nine acceptance criteria were met, three were not met, and two were not assessed.

Objective (i) was met, with the two source substitutions documented in Table 4.1. Objective (ii) was met in full. Objective (iii) was met in respect of map construction but not in respect of validation performance. Objective (iv) was met functionally but not architecturally. Objective (v) was partially met.

The P-factor and R-factor criteria were not assessed because calibration was performed through direct parameter manipulation rather than through SUFI-2, and no uncertainty ensemble was generated.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

In this chapter, the conclusions drawn from the hydrological modelling, flood susceptibility mapping and validation of both components are presented, together with recommendations for improving model performance and for the use of the outputs by stakeholders.

The primary aim of this study was to design and build an interactive web GIS portal for flood risk analysis in Lagos State, supported by SWAT+ hydrological simulation and AHP-based flood susceptibility mapping, and to validate both components against defined acceptance criteria. The research aimed to establish whether a process-based hydrological model could be calibrated for the Ogun catchment in the absence of gauged discharge, and whether the resulting susceptibility map corresponded to observed flooding.

Findings from the study show that the SWAT+ model met all three of its acceptance criteria, achieving NSE of 0.507, R^2 of 0.606 and PBIAS of +1.6% over the full evaluation period, with validation-period performance exceeding calibration-period performance. Catchment discharge was found to respond non-linearly to increased rainfall, with a 10% rainfall increase producing a 15.5% increase in mean discharge and a 30% increase producing 47.6%.

For the flood susceptibility map, the results were mixed. The map ranks flood susceptibility correctly, with observed flooding enriched monotonically across the five zones and reaching 1.61 times chance expectation in the Very High class. However, the map met only one of its four acceptance criteria. AUC-ROC of 0.62, sensitivity of 0.685 and overall accuracy of 0.526 all fall below their thresholds, and the map therefore possesses ordinal but not classificatory skill.

An estimated 4,810,069 residents, representing 38.7% of the population of Lagos State, were found to occupy High or Very High susceptibility zones.

5.1 Discussion

While the study demonstrates that a SWAT+ model can be calibrated to acceptable standards using modelled rather than gauged discharge, the results reveal two limitations that constrain the strength of the conclusions.

The first concerns the calibration reference. Because GEOGLOWS is itself a model output, the performance statistics reported in Table 4.6 measure agreement between two models rather than agreement with observation. This approach is recognised for ungauged and poorly gauged basins and was adopted deliberately following Odusanya et al. (2019), but it cannot support the same strength of claim as calibration against gauged discharge. The independent evapotranspiration validation reported in Section 4.6 partially offsets this constraint, since it introduces both a different variable and a data source unconnected to GEOGLOWS, one of which is a direct satellite retrieval. That the model reproduces annual evapotranspiration to within 14.4% without ever having been fitted to it provides evidence of physical realism that the discharge statistics alone cannot supply.

The corresponding failure to reproduce seasonal evapotranspiration amplitude, reported in Table 4.12, is the model's clearest identified weakness. The diagnosis given in Section 4.6.3 is that the PET multiplier is constrained at its permitted maximum of 1.20, while both reference products independently indicate that simulated evapotranspiration is 12 to 14% too low. Two separate lines of evidence therefore converge on the conclusion that the model requires more evapotranspiration than the parameter range permits. Relaxing that constraint improves evapotranspiration but drives discharge performance below the acceptance thresholds, so no compromise within the tested parameter space satisfies both.

The second limitation concerns the susceptibility map's discriminatory performance. The contrast between Table 4.20 and Table 4.21 is central: enrichment rises monotonically across all five zones, yet AUC-ROC reaches only 0.62 and specificity barely exceeds chance. These results are consistent rather than contradictory. The map carries genuine information about relative susceptibility while

lacking the sharpness to classify individual locations, because the High and Very High zones together occupy nearly half the state.

Three differences in validation design are relevant when comparing the AUC of 0.62 obtained here with values above 0.80 commonly reported in the flood susceptibility literature. First, many published studies validate against inventories of flood points that were themselves used, in whole or in part, to inform factor weighting, which inflates apparent performance. The present validation used an independent satellite-derived flood extent that played no role in constructing the map. Second, validating against a continuous 35.2 km² flood extent is a more demanding test than validating against a small set of discrete points, as it requires discrimination across every cell of the observed footprint. Third, AHP weights were assigned from expert judgement without calibration against observed flooding, whereas data-driven approaches such as random forest or logistic regression fit weights to the inventory directly and would be expected to score higher against it. The lower AUC therefore reflects a stricter validation design as well as genuine model limitations, and these two contributions cannot be separated from the present results alone.

A further observation arises from Section 4.8.4. The AHP weighting reported in Table 4.14 assigns 0.75 of the index to terrain-related factors, all of which describe topographic position. Flooding caused by insufficient or obstructed drainage in built-up areas is largely independent of terrain position at the 30 m scale and is represented only through the land cover factor, which carries a weight of 0.05 and derives from a classification with approximately 53% accuracy. Two independent observations are consistent with this: the negative rank correlation of −0.45 in Table 4.23, and the low modelled exposure of Alimosho, Agege and Ifako/Ijaye in Table 4.24 despite their high reported flood incidence. Neither observation is conclusive, since both derive from a dataset shown to be confounded by population, but the possibility that the index systematically under-represents urban pluvial flooding warrants investigation.

Additional constraints on the study should be noted. Oyan Dam lies within the modelled basin and its releases are not represented, placing the 2022 flood outside the model's scope. Soils were represented by a single topsoil texture layer, with 88% of the basin assigned to one class. The 2012 MODIS flood extent has a resolution of 250 m, which cannot resolve street-scale urban flooding, so non-detection is not evidence of no flooding. The susceptibility map is a static surface validated against a single flood season. Finally, P-factor and R-factor were not computed, as no uncertainty ensemble was generated.

5.2 Conclusion

The following overall conclusions can be drawn from the results of this study:

1. A SWAT+ model of the Ogun catchment calibrated against modelled discharge achieves satisfactory monthly performance, with full-period NSE of 0.507, R^2 of 0.606 and PBIAS of +1.6%. Validation-period performance exceeds calibration-period performance, indicating the parameter set is not over-fitted. The margins above threshold are narrow, and the model should be regarded as satisfactory rather than robust.
2. Catchment discharge responds non-linearly to increased rainfall, with an amplification factor of approximately 1.55. Linear scaling of rainfall therefore understates the hydrological consequences of intensifying rainfall, which supports the inclusion of a process-based component alongside index-based mapping.
3. Independent validation against two unrelated evapotranspiration products, one of them a direct satellite retrieval, shows that the model reproduces annual evapotranspiration volume to within 14.4% while underestimating its seasonal amplitude by approximately a factor of three. As the model was never calibrated to evapotranspiration, this constitutes genuine out-of-sample evidence of partial physical realism together with a specific identified deficiency.

4. Improving simulated evapotranspiration and maintaining discharge performance are mutually incompatible within the available parameter space, as the PET multiplier is constrained at its permitted maximum.
5. The AHP susceptibility map ranks flood susceptibility correctly. Observed flooding is enriched monotonically across the five zones, reaching 1.61 times chance expectation in the Very High zone, and 67.08% of observed flood extent falls within the High and Very High zones against 48.91% of the state area.
6. The same map fails three of its four acceptance criteria, with AUC-ROC of 0.62, sensitivity of 0.685 and overall accuracy of 0.526. The map possesses ordinal but not classificatory skill, and Objective (v) is therefore only partially attained.
7. An estimated 4,810,069 residents, or 38.7% of the population of Lagos State, occupy High or Very High susceptibility zones, with LGA-level exposure ranging from 67.8% in Eti-Osa to 7.2% in Ifako/Ijaye. This estimate is conditional on the classification limitations stated in conclusion 6.
8. Agency flood-report counts are unsuitable as validation data for susceptibility mapping in this setting, as NEMA 2022 records correlate more strongly with raw population, at ρ of +0.57, than with modelled exposure, at ρ of +0.17.
9. A functional decision-support portal integrating hazard, exposure, scenario and validation views can be delivered on a lightweight single-process architecture, without the spatial database and tile-server stack originally specified.

5.3 Recommendations

Based on the findings of this study, the following recommendations are made:

1. The susceptibility map should be used for relative prioritisation between areas and not for site-specific determinations. The portal's framing as a susceptibility and scenario tool rather

than a forecasting instrument should be preserved in any published version, as this is consistent with what the validation demonstrates.

2. Flood incident reporting by LASEMA, NEMA and allied agencies should record coordinates rather than community names alone. The principal obstacle to validating flood models for Lagos is the absence of a geolocated flood inventory, and Section 4.8.4 quantifies the consequence: an otherwise valuable 62-record dataset could support only LGA-level analysis, which proved confounded by population.
3. Drainage and reservoir planning should account for the non-linear discharge response quantified in Table 4.8, in which a 30% rainfall increase produces a 47.6% increase in mean discharge and a 28.2% increase in peak-month discharge at the Ogun outlet.
4. The proposition that the index under-represents urban pluvial flooding should be tested by repeating the overlay analysis with the susceptibility surface stratified by built-up density. If confirmed, a drainage-related factor such as impervious fraction or drainage density should enter the index at a weight materially higher than the 0.05 currently assigned to land cover.
5. Logistic regression or random forest models should be fitted to the same conditioning factors and the same 2012 flood extent, in order to separate the contribution of AHP's uncalibrated weighting from the strictness of the validation design.
6. SUFI-2 should be run over the calibrated parameter set to produce the P-factor and R-factor statistics left unassessed in Table 4.26 and to place confidence bounds on the scenario results.
7. Oyan Dam operation should be incorporated into the model, which would bring the 2022 event within its scope and improve its value for flood early warning, since release-driven events are those for which advance notice is most achievable.
8. Sentinel-1 SAR imagery, offering 10 m all-weather flood mapping, should be used to compile flood extents for several seasons. This would permit validation at a scale appropriate to urban flooding and replace the present single-event assessment.

9. The source of the evapotranspiration seasonality deficiency should be investigated to establish whether it originates in the CHIRPS rainfall forcing, the single-layer soil representation, or the PET formulation, using multi-variable calibration with a widened PET multiplier range.

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